
Supplementary Information for: Bridging the Gap Between Scientific Laws Derived by AI Systems and Canonical Knowledge via Abductive Inference with AI-Noether

Karan Srivastava

University of Wisconsin-Madison || IBM Research
Madison, WI || Yorktown Heights, NY
ksrivastava4@wisc.edu

Sanjeeb Dash

IBM Research
Yorktown Heights, NY
sanjeebd@us.ibm.com

Ryan Cory-Wright

Imperial Business School
London, UK
r.cory-wright@imperial.ac.uk

Barry Trager

IBM Research
Yorktown Heights, NY
bmt@us.ibm.com

Cristina Cornelio

Samsung AI
Cambridge, UK
c.cornelio@samsung.com

Lior Horesh

IBM Research
Yorktown Heights, NY
lhoresh@us.ibm.com

Supplementary Note 1: Extended Literature Review

Abductive reasoning is a field of inference that generates plausible explanations from incomplete observations [53, 51]. Unlike deductive reasoning, which obtains a conclusion from a set of premises, abductive reasoning starts with an outcome and ends with a set of plausible explanations, e.g.,

- The fact C is observed.
- If A were true then C would be true.
- There is reason to believe that A is true.

As observed by Peirce [53], abductive reasoning is arguably “the only logical operation which introduces any new ideas”. Moreover, it is often applied by humans in everyday situations, such as reading between the lines [50] and counterfactual reasoning [52]. Doctors use abductive reasoning to propose possible diseases that cause observed symptoms. It has been argued that automating inference may be necessary for understanding and automating consciousness [8]. However, abductive reasoning has not yet been incorporated into state-of-the-art AI systems, such as AI-Feynman, AI-Descartes, or AI-Hilbert [67, 12, 13]. Thus, automating abductive inference is an important component of automating the scientific method.

Automated Abductive Inference in Logic and Classical AI. Abduction, firstly introduced by Peirce as the inferential step that proposes an explanatory hypothesis for a surprising observation in the logic of scientific inquiry [19], was then later on, in AI, sharpened into a logic-based hypothesis synthesis problem [46]. Given an incomplete background theory T and observations (target claim) O , find hypotheses H (additional assumptions from a designated space of abducibles, e.g., defined by a grammar) that render the claim derivable, meaning that $T \cup H \models O$ and $T \cup H$ is consistent, typically subject to minimality criteria [26] ($\models$ is the standard logical entailment symbol, meaning that O is true in every model satisfying $T \cup H$). This was first introduced for propositional logic and then extended to more expressive forms such as first-order [46]. Abductive logic programming

(ALP) then further operationalized these ideas by combining abducibles with integrity constraints and proof procedures [38], and more recent work extended it to even richer constraint families (e.g., including probabilistic integrity constraints [7]).

Most relevant to our work is the case in which hypotheses are *equations* (or more generally, algebraic constraints): in this setting, abduction can be realized via consequence generation in equational logic, where explanations are returned as equalities/disequalities that complete the theory enough to derive the target [23]. More broadly, implicate-generation modulo background theories can be used to generate missing conditions (including equalities and arithmetic constraints) whenever a suitable decision procedure is available [21, 22]. However, these methods either support only limited arithmetic expressiveness (usually including only basic arithmetic operators) or are computationally challenging as the hypothesis space grows.

More recently, abduction has become part of automated reasoners/solvers (see the next paragraph for more details): SMT/SyGuS-based methods synthesize hypotheses directly in background theories with equality or linear arithmetic [57]. Automated solvers, such as *cvc5*, expose abduction as a reusable primitive [2]. These can also be incorporated into verification-oriented systems where abduction fills in preconditions and specifications so that program proofs can find a proof compositionally [65, 3].

Finally, beyond logic, several papers have investigated abductive inference in the context of scientific discovery: Swanson and Smalheiser’s interactive system ARROWSMITH [66] for identifying complementary literature in other branches of science exploits abductive hypothesis formation as constructing bridging concepts across disjoint research areas; and more recently Langley [40] argued for integrated discovery systems that treat abduction as one component within an end-to-end pipeline that combines multiple reasoning and learning mechanisms.

We follow this line of work in interpreting abduction as the process of selecting or synthesizing hypotheses that complete an incomplete theory in order to explain a target hypothesis.

Automated Abductive Inference for Machine Learning. The recent surge in popularity of black-box models, including large language models, has led to an interest in explaining the predictions of these models. LIME [58] and similar methods such as SHAP [45] provide (local) interpretable explanations that describe the local behavior of any ML model using a weighted linear combination (often sparse) of input features; see [60, 42] for reviews of interpretable and explainable machine learning. However, existing works on interpretability and explainability generally focus on computing explanations for black-box models, rather than generating theories that could be used for scientific discovery. In particular, techniques like LIME or SHAP scores cannot, to our knowledge, be leveraged to establish missing axioms in a scientific discovery context.

More recently, a new line of work has explored abductive inference for machine learning [16, 37, 73, 36, 72]; see also [69] for a recent review of this line of work. Initiated by [16, 37], the key idea is to embed purely logical reasoning models within the machine learning process to take advantage of the benefits of both machine learning and logical reasoning. For instance, in [37], the authors use abductive inference to generate explanations of ML models given input background theories and a set of hypothetical explanations. [16] combines machine learning techniques with a first-order logic model to obtain an output that depends on both models. However, these works do not apply abductive inference to scientific discovery.

SMT Solvers and Abductive Reasoning in Automated Theorem Proving. Recent work has explored the use of Satisfiability Modulo Theories (SMT) solvers for abductive reasoning, particularly in the context of automated theorem proving and program verification. SMT solvers generalize Boolean SAT by incorporating theories over specific domains (e.g., arithmetic, arrays, bit-vectors), enabling reasoning about richer constraints [4, 18].

Barbosa et al. [3] developed an SMT-based tactic in the Coq proof assistant that uses abductive reasoning to identify missing axioms needed to complete a proof. When the *cvc5* SMT solver fails to prove a goal, their *abduce* tactic generates facts that, when added to the background theory, allow the proof to succeed. This work shares conceptual similarities with AI-Noether, as both identify missing axioms to bridge a derivability gap, though their approach operates within formal proof systems rather than scientific discovery contexts. Reynolds et al. [57] introduced scalable algorithms for abduction via syntax-guided synthesis (SyGuS), demonstrating that SMT solvers can efficiently

generate formulas consistent with axioms that entail a given goal. Their work establishes theoretical foundations for abductive synthesis that inform our comparison with `cvc5` in Supplementary Note 6.

Beyond theorem proving, abductive reasoning has been applied to scientific hypothesis generation through Abductive Logic Programming (ALP). For example, Ray [56] demonstrated how ALP methods outperform theorem-proving approaches. While ALP focuses on logic-based hypothesis synthesis, AI-Noether operates in the polynomial algebraic setting, leveraging computational algebraic geometry rather than first-order logic, and works in the robust cases of noisy machine-generated hypotheses.

Large Language Models for Scientific Discovery and Equation Discovery. The emergence of large language models (LLMs) has opened new avenues for automating scientific discovery. Recent work has explored using LLMs’ embedded scientific knowledge and code generation capabilities to discover governing equations from data, a task traditionally addressed by symbolic regression.

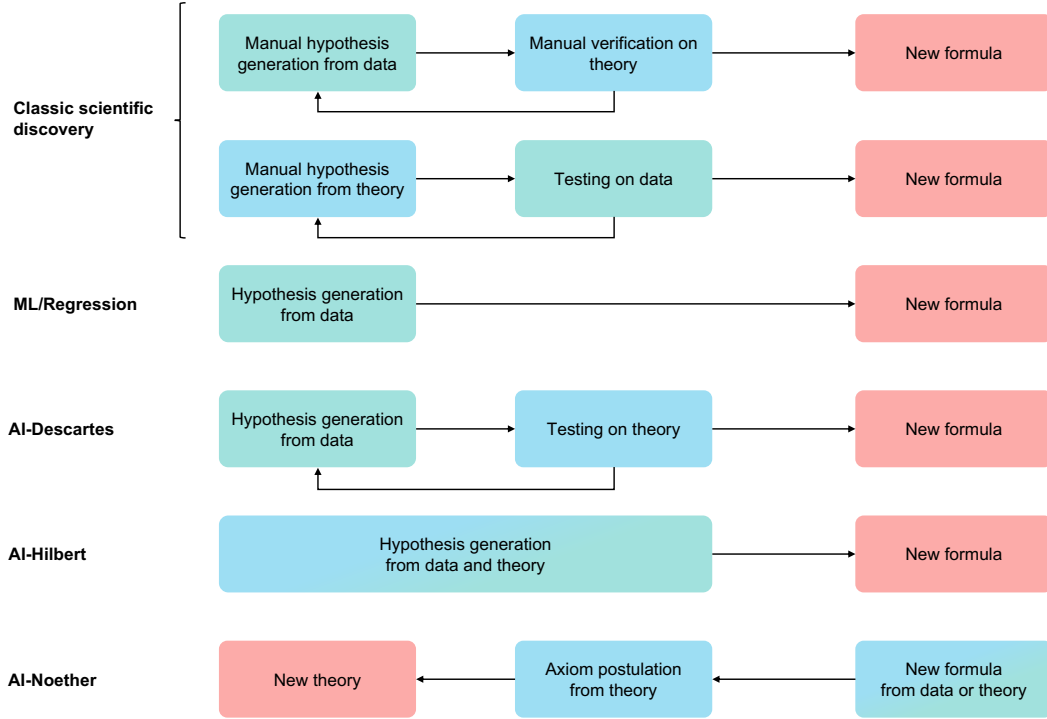
Shojaee et al. [61] introduced LLM-SR, which combines LLMs with evolutionary search to discover scientific equations by treating them as program skeletons with placeholder parameters. Their approach leverages domain-specific priors from LLM training to navigate the equation search space more efficiently than classical symbolic regression methods. Follow-up work by DrSR [70] enhances this framework with dual reasoning that combines data-driven insights with reflective learning, analyzing structural relationships like monotonicity and correlation to guide equation generation. These LLM-based symbolic regression approaches are complementary to AI-Noether: while they discover equations *from data*, AI-Noether discovers equations needed to *explain* already-formulated hypotheses without requiring data for the missing axioms.

Beyond symbolic discovery, LLMs have enabled autonomous experimental systems. Boiko et al. [9] demonstrated fully autonomous chemical research where LLMs plan multi-step syntheses, control laboratory robotics, and analyze results - closing the loop between computational hypothesis generation and physical experimentation. This represents a step toward Kitano’s “Nobel Turing Challenge” [39] of building AI systems capable of Nobel-quality autonomous discovery.

More generally, LLMs have been applied to automated theorem proving and mathematical discovery. Romera-Paredes et al. [59] introduced FunSearch, an evolutionary procedure pairing pretrained LLMs with systematic evaluation to solve open problems in extremal combinatorics, demonstrating that LLMs can push beyond existing human knowledge when combined with search algorithms. Lu et al. [44] proposed The AI Scientist, a system for fully automated open-ended scientific discovery that generates research ideas, runs experiments, and writes papers. In formal mathematics, recent work on automated theorem proving using LLMs [55, 71] has shown promise in completing proof steps, though performance still lags human experts on research-level problems. These systems focus on *generating* new knowledge or proofs, whereas AI-Noether performs *abductive inference* to identify which axioms are missing from an incomplete theory. The axioms we are inferring are, by assumption, not derivable from the other known axioms and consequences, so traditional forward inference approaches would not work.

Neuro-Symbolic Methods and Physics-Informed Discovery. Recent work has explored hybrid approaches that combine neural networks with symbolic reasoning for scientific discovery. Petersen et al. [54] introduced deep symbolic regression using reinforcement learning with risk-seeking policies to discover mathematical expressions, demonstrating improved exploration of the symbolic search space. Valipour et al. [68] developed SymbolicGPT, a transformer-based model for symbolic regression that learns from large corpora of synthetic equations. These neural-symbolic approaches excel at pattern recognition and generalization across equation families, but typically require training data for the target expressions.

Physics-informed approaches incorporate domain knowledge as inductive biases. Gao et al. [27] showed that embedding physical constraints (e.g., conservation laws, symmetries) into neural network architectures improves equation discovery accuracy and generalization. Similarly, SINDy [10] leverages sparsity constraints to identify governing equations of dynamical systems, and Hamzi and Owadi [31] similarly demonstrate the use of kernel flows in learning dynamical systems. While these methods successfully integrate prior knowledge, they focus on *parameter fitting* within known functional forms rather than *structural discovery* of missing axioms. AI-Noether addresses the complementary problem where the functional form itself is unknown and must be inferred from algebraic constraints.



Supplementary Fig. 1: **Comparison of AI-Noether with other scientific discovery models.** We compare classical scientific discovery, machine-learning-based regression, and recent integrated systems (AI-Descartes and AI-Hilbert) with AI-Noether. While most approaches focus on generating new hypotheses or formulas from data and/or existing theory, AI-Noether targets hypotheses that are not supported by the current theory and performs abductive inference to identify and justify the missing axioms required to reconcile them with known knowledge.

AI-Noether vs SOTA Scientific Discovery Models. Supplementary Fig. 1 compares AI-Noether with four other scientific discovery models in terms of their functionality in discovering theory from data and existing theory. The traditional discovery process starts with known theory and data and seeks to discover a new formula that explains the data in a way that is consistent with the theory. The traditional, first-principles-based scientific discovery, broadly speaking, operates on two paradigms: either generate a hypothesis by studying patterns in known data and verify the hypothesis by checking derivability or consistency with theory or derive possible hypotheses from first-principles and test the derived hypotheses on data. This approach is manual and time-consuming. ML/Regression approaches help automate the hypothesis generation step by allowing for automated hypothesis generation using data. Verification here, however, is still done manually and is often the bottleneck. AI-Descartes and AI-Hilbert solve this problem by automating the verification step by either automating the testing on theory using theorem provers (AI-Descartes) or combining fit-to-theory in the search process (AI-Hilbert). AI-Noether targets a new problem of automation in the scientific process: in the event that theory is insufficient for deriving a machine-generated hypothesis, instead of changing the hypothesis-generation mechanism, AI-Noether proposes corrections / completions to the theory in order to discover a new theory required to validate the hypothesis.

AI-Noether inside a broader scientific discovery loop. AI-Noether is complementary to systems such as AI-Feynman, AI-Descartes, AI-Hilbert, symbolic-regression methods, and broader AI Scientist-style pipelines. While these systems primarily generate, fit, or validate candidate laws and hypotheses, AI-Noether addresses the distinct theory-discovery task: inferring missing axioms that make a candidate hypothesis derivable from an incomplete background theory.

Expanding on the end-to-end view of Fig. 2 introduced in the main text, candidate hypotheses may be generated by upstream tools, such as symbolic-regression systems, AI-Descartes, AI-Hilbert, LLM-based discovery agents, or by human scientists. AI-Noether then takes such a hypothesis together with the current background theory and performs abductive theory formation, producing

candidate missing axioms that explain why the hypothesis should follow. Other components can further support this loop: hypotheses may be reranked according to numerical fit [67], theoretical consistency [12], simplicity [13], or derivability [12]; verifiers such as theorem provers [12], or algebraic proof systems [13] can test whether a hypothesis follows from the background theory; counterexample or falsification modules [12] can search for contradictions or non-generalizing hypotheses; and experimental-design modules [35] can propose new data-collection conditions when additional evidence is needed.

Together, these components constitute two complementary modes of discovery: the first is the discovery of new phenomena or candidate laws, which may be produced by experiments, human scientists, or upstream AI systems; the second is the discovery of new theory: the identification of axioms or principles that explain why such laws hold and how they may generalize beyond the original phenomenon. AI-Noether is thus not intended to replace the full scientific discovery pipeline; rather, it automates this second mode of discovery by inferring missing axioms that complete, explain, or generalize the environment in which the observed phenomenon occurs.

Supplementary Note 2: A Walk-Through Example

Example 1. Consider an axiom system over five real quantities, indicated by x, y, z, a, b :

$$x^2 - ay^2 = 0,$$

$$by - z = 0.$$

Further, suppose that a, b are physical constants in the system, while x, y, z are quantities that may vary. Let p_1 be the polynomial $x^2 - ay^2$ and let $p_2 = by - z$, and the axioms be $p_1 = 0$ and $p_2 = 0$. Further, let

$$q = b^2x^2 - az^2.$$

The equation $q = 0$ then follows from the axioms as

$$q = b^2x^2 - az^2 = b^2(x^2 - ay^2) + a(by + z)(by - z) = b^2p_1 + a(by + z)p_2. \quad (\text{S1})$$

In the example above, we have expressed q as $\alpha p_1 + \beta p_2$ where $\alpha = b^2$ and $\beta = a(by + z)$ are polynomial multipliers in the same polynomial ring $\mathbb{R}[x, y, z, a, b]$. We say that q is an *algebraic combination* of the polynomials p_1 and p_2 and say that $q = 0$ is *derivable* from $p_1 = 0$ and $p_2 = 0$. As $q = \alpha p_1 + \beta p_2$, it follows that $q = 0$ for all values of x, y, z, a, b such that $p_1 = 0$ and $p_2 = 0$. We therefore say that the axioms in the example imply or explain $q = 0$.

Supplementary Table 1: Six minimal axiom systems; each column corresponds to one such system, consisting of two equations that explain $q = 0$.

$b = 0$	$b = 0$	$b = 0$	$x^2 - ay^2 = 0$	$x^2 - ay^2 = 0$	$x^2 - ay^2 = 0$
$a = 0$	$by + z = 0$	$by - z = 0$	$a = 0$	$by + z = 0$	$by - z = 0$

Notice that there are many alternative theories from which the equation $q = 0$ follows as a consequence. For example, looking at the equation (S1) and letting $p_3 = by + z$, it is clear that $q = 0$ follows from the axioms $p_1 = 0$ and $p_3 = 0$ as $q = \alpha p_1 + \beta p_3$. In a similar manner, we can trivially create the six *minimal* theories in Supplementary Table 1 where the first axiom is formed by taking a polynomial that is a factor of αp_1 and setting it to 0, and the second axiom is similarly composed from a factor of βp_2 ; $q = 0$ is a consequence of each theory.

Each of the 6 axiom systems above is minimal in the sense that we cannot remove an axiom from it and still have $q = 0$ as a consequence. Depending on the context, some axiom systems may be more interesting than others. For example, if a represents the mass of a pendulum, $a = 0$ is likely not an interesting condition. On the other hand, if a represents temperature in Celsius, $a = 0$ may represent a meaningful condition.

The axiom system $p_1 = 0, p_2 p_3 = 0$ is also a valid theory explaining $q = 0$, but we call it *reducible* in the sense that we can trivially get a simpler axiom system if we replace $p_2 p_3$ by either of its factors. Thus, if our only available axioms were $p_1 = 0$, the “residual” of the set of axioms $\{p_2 p_3 = 0\}$ is not as small as possible. There are other ways of building an infinitude of axiom systems such that $q = 0$ is a consequence. Starting from the axiom system $p_1 = 0, p_2 = 0$, it is easy to see that

$p_1 = 0, rp_1 + sp_2 = 0$, where r, s are nonzero real numbers, are valid theories that explain $q = 0$. Again, starting with $p_1 = 0$, the axiom set $\{rp_1 + sp_2 = 0\}$ does not have the least residual.

Now, assume we have experimental data for a phenomenon consisting of a number of data points specifying values of x, z, a, b (with a and b unchanging in the data), that the data satisfy the equation $b^2x^2 - az^2 = 0$, and that we are able to numerically obtain this equation from the data. Further, suppose that we have a partial theoretical explanation of the phenomenon, where the partial theory assumes the presence of another variable y and the equation $x^2 - ay^2 = 0$, i.e., $p_1 = 0$. In this situation, we consider y as an unmeasured quantity, and x, z, a , and b as measured quantities. Given that the data points at hand do not have any values for y , it is not possible to infer the relationship $by - z$ directly from the data.

Consider the equation $0 = b^2p_1 = b^2x^2 - b^2ay^2$ that follows from $p_1 = 0$. Subtracting the derived equation $b^2x^2 - az^2 = 0$ from it, we obtain the relationship $a(z^2 - b^2y^2) = 0$ as a consequence of $p_1 = 0$ and $q = 0$. This implies that either $a = 0$ or $by - z = 0$ or $by + z = 0$. Each of the equations (or products of pairs of equations) in the previous line can be appended to $p_1 = 0$ to get a valid theory (in the absence of other information) explaining $q = 0$.

Using the notation above, “derivable from” implies “can be explained by”. We will later discuss conditions under which the converse is true (see the Criteria for Recoverability of Axioms subsection of the Methods in the main text).

If $q' = bx^2 - ayz - 1$, then the expression $q' = 0$ is inconsistent with the axioms $p_1 = 0$ and $p_2 = 0$; this follows from the fact that $bp_1 + ayp_2 = bx^2 - ayz = q' + 1$ and therefore $1 = 0$ (a trivially false statement) is derivable from $p_1 = 0, p_2 = 0$ and $q' = 0$. In general, it is not always true that $1 = 0$ can be derived from some axioms and an inconsistent hypothesis, but similar false expressions can always be derived (e.g., $1 \leq 0$). See Supplementary Note 3 for more details on how we test for inconsistency.

Supplementary Note 3: Methodology Details

In this section, we develop the details of *AI-Noether* and its **Encode-Decompose-Reason** modules. We first develop the algebraic theory of the method and then extend it to numerical algorithms that enable us to handle noise. For further reading on these topics, one can refer to the following textbooks and resources on computational algebraic geometry [14, 62, 1, 24, 28, 25, 32]. This section will include all the algebraic geometry framework and, as such, will be denser than the rest of the paper.

3.1 Preliminaries - Algebraic Background Without Noise

We define notions specific to real algebraic geometry; see [14] for further details. Let $\mathbb{R}$ denote the set of real numbers and let $R = \mathbb{R}[x_1, \dots, x_n]$ be the ring of polynomials defined over the n variables $x_1, \dots, x_n$ with real coefficients. Axioms and hypotheses in our model will be equations of the form $p(\mathbf{x}) = 0$ where $p \in R$. We abuse notation and identify an axiom by its defining polynomial. We will focus on real-valued solutions of axiom systems; see Supplementary Note 5 for more details. Finally, we will focus the exposition on the case with one axiom correction and one consequence for ease of explanation. This will generalize well to the multiple missing axiom and consequence case and we will make a note of these generalizations for each module.

Encode. An *ideal* over R is a subset I of R that is closed under addition and also multiplication by elements of R : if $f, g \in I$, then $f + g \in I$ and $\alpha f \in I$ for any $\alpha \in R$. Given polynomials $A_1, \dots, A_k \in R$, we denote the set of all algebraic combinations of $A_1, \dots, A_k$ by

$$\langle A_1, \dots, A_k \rangle = \{f \in R : f = \sum_{i=1}^k \alpha_i A_i \text{ for some } \alpha_i \in R\}.$$

This set is called an ideal over R with *generators* $A_1, \dots, A_k$. Let $\mathcal{A}$ stand for $\{A_1, \dots, A_k\}$, and let $I(\mathcal{A})$ stand for the ideal $\langle A_1, \dots, A_k \rangle$. Our earlier notion of “derivability” corresponds to ideal membership: $Q = 0$ is derivable from $\mathcal{A}$ if Q is an algebraic combination of $A_1, \dots, A_k$, i.e., if $Q \in I(\mathcal{A})$. Given an ideal I , we define the *variety* of I , denoted $V(I)$, as the set

$$V(I) = \{\mathbf{x} \in \mathbb{R}^n : f(\mathbf{x}) = 0 \text{ for each } f \in I\}.$$

A standard fact from algebraic geometry is that the set above is the same as the solution set to any finite collection of polynomials that generate the same ideal. Encoding equations with ideals and varieties has a key advantage in that it allows us to study logical implication and consistency questions geometrically: if $Q = 0$ can be derived from $A_1 = 0, \dots, A_k = 0$, then $V(A_1, \dots, A_k, Q) = V(A_1, \dots, A_k)$. If adding Q to $\mathcal{A}$ introduces extra structure into the resulting variety, then Q is not derivable from $\mathcal{A}$. As such, this holds for any Q , and so in the case where we have multiple consequences Q_i , this holds for each Q_i . In that case, we define $V(A_1, \dots, A_k, Q_1, \dots, Q_s)$ as our variety.

Decompose. Assume Q is not implied by $\mathcal{A}$. Then $V(\mathcal{A}, Q) \neq V(\mathcal{A})$. We will detail this in the case where a single axiom correction is needed. Suppose Q can be derived *nontrivially* from an axiom A_{k+1} along with $\mathcal{A}$, i.e., $Q = \alpha_1 A_1 + \dots + \alpha_{k+1} A_{k+1}$ for some polynomials α_i without $\alpha_{k+1} = 1, A_{k+1} = Q$. In this case $V(A_1, \dots, A_k, Q) = V(A_1, \dots, A_k, \alpha_{k+1} A_{k+1})$ and since $\alpha_{k+1} \neq 1$, we know that intersecting $V(Q)$ with $V(\mathcal{A})$ is the same as intersecting $V(\alpha_{k+1} A_{k+1})$ with $V(\mathcal{A})$. Since $V(\alpha_{k+1} A_{k+1})$ is reducible, we are introducing new reducibility into $V(\mathcal{A})$ that is a factor of the residual $\alpha_{k+1} A_{k+1} = Q - (\sum_{i=1}^k \alpha_i A_i)$ (and if we had multiple consequences, then we would have $V(A_1, \dots, A_k, Q_1, \dots, Q_s) = V(A_1, \dots, A_k, \alpha_{k+1} A_{k+1}, \dots, \alpha_{l-k+1} A_{l-k+1})$). We therefore can take our intersected variety $V(\mathcal{A}, Q)$ and study its irreducible components. We next define the mechanisms required for this process.

A *prime ideal* is an ideal $P \subset R$ such that if $fg \in P$ with $f, g \in R$, then $f \in P$ or $g \in P$.

For a prime ideal P , $V(P)$ is irreducible, i.e., it cannot be represented as the union of two proper subsets that are varieties. The *radical* of an ideal I is denoted by $\sqrt{I}$ and is defined as:

$$\sqrt{I} = \{f \in R : f^m \in I \text{ for some positive integer } m\}.$$

The following key theorem allows us to break up a variety into irreducible components:

Theorem 1: Primary Decomposition

Any ideal I can be decomposed into an intersection of ideals

$$I = I_1 \cap \dots \cap I_r$$

such that

1. There are no redundancies among the I_i , i.e., for each j we have $\cap_{i \neq j} I_i \not\subset I_j$.
2. For each i , $\sqrt{I_i}$ is a prime ideal (called an *associated prime* of I).

The proof of this can be found in [1, 14, 32, 24].

Let $P_i = \sqrt{I_i}$. Using the fact that for any ideals I, J , we have $V(I \cap J) = V(I) \cup V(J)$ and $V(I) = V(\sqrt{I})$ [14], we get from the primary decomposition that a variety can be decomposed into a union of varieties corresponding to prime ideals: $V(I) = \bigcup_{i=1}^r V(P_i)$.

Decomposition in this setting is analogous in spirit to Principal Component Analysis (PCA) in linear algebra, identifying latent, independent substructures that explain variation in the system. Whereas PCA decomposes a dataset into orthogonal directions of variance, primary decomposition identifies algebraically independent pieces of a solution space introduced by inconsistency or incompleteness.

Reason. To test derivability, we rely on an algebraic technique for eliminating variables from polynomial systems. A *reduced Gröbner basis* is a canonical generating set for an ideal that can be computed algorithmically. Given a variable ordering $x_1 < \dots < x_n$ (e.g., lexicographic), the reduced Gröbner basis $\{G_1, \dots, G_m\}$ of $I = \langle A_1, \dots, A_k \rangle$ satisfies a key elimination property:

Theorem 2: Elimination Theorem

For any $1 \leq d \leq n$, the polynomials in the Gröbner basis $\{G_1, \dots, G_m\}$ involving only $x_1, \dots, x_d$ generate the same ideal as all polynomials in I involving only those variables:

$$\langle \{G_1, \dots, G_m\} \cap \mathbb{R}[x_1, \dots, x_d] \rangle = \langle A_1, \dots, A_k \rangle \cap \mathbb{R}[x_1, \dots, x_d].$$

The proof of this can be found in [1, 14, 32, 24].

So if we want to eliminate the variables $x_{d+1}, \dots, x_n$ from the system, we compute a set of generators for the elimination ideal $I \cap \mathbb{R}[x_1, \dots, x_d]$ by finding the reduced Gröbner basis and extracting polynomials involving only $x_1, \dots, x_d$. Geometrically, this corresponds to projecting the variety $V(I) \subset \mathbb{R}^n$ onto the subspace of the first d coordinates. The elimination ideal defines the Zariski closure of this projection: $V(I \cap \mathbb{R}[x_1, \dots, x_d]) = \overline{\pi_d(V(I))}$, where the closure is taken over $\mathbb{C}$ and we then restrict to real points.

We note that Gröbner basis computation has doubly exponential worst-case complexity in general [48], although in our benchmark systems the structured and relatively sparse nature of the polynomial ideals considered makes our computations tractable.

This provides a way to test algebraic consequences: if Q depends only on $x_1, \dots, x_d$, we eliminate all other variables and check whether Q belongs to the elimination ideal. We use this to test whether Q can be derived from the axioms together with a candidate axiom extracted from decomposition. Elimination also filters out trivial candidates whose projections represent undesired boundary conditions under which Q holds vacuously.

We present sufficient conditions for axiom recovery in the Criteria for Recoverability of Axioms subsection of the Methods in the main text. The generalized algorithm is detailed in Algorithm 1 in the Algorithm Implementation Details subsection of Supplementary Note 5.

Our missing axiom construction attempts to find axioms that are sufficient to derive Q , i.e., to decide whether or not Q is contained in the ideal generated by the axioms. If we want to decide whether or not a set of axioms implies Q , given the ideal I generated by the axioms, we are asking whether Q vanishes on all points of $V(I)$. To make this test effective, we need to construct the biggest ideal with the same set of zeros, i.e., the set of all polynomials which vanish on $V(I)$. Since the points of $V(I)$ are real, the set of all polynomials which vanish on $V(I)$ can be computed as the real radical of the ideal I . [49].

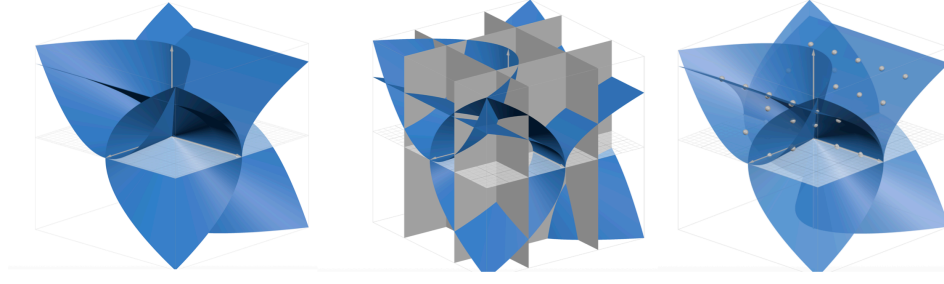
3.2 Handling Noisy Consequences - Numerical Extensions

The purpose of this study is to bridge the gap between machine-generated hypotheses and theory. Therefore, we must address the case where consequences (and possibly axioms) contain noise due to measurement error or numerical approximation in data-driven discovery. Accordingly, we make two key changes to our algorithm to account for noise.

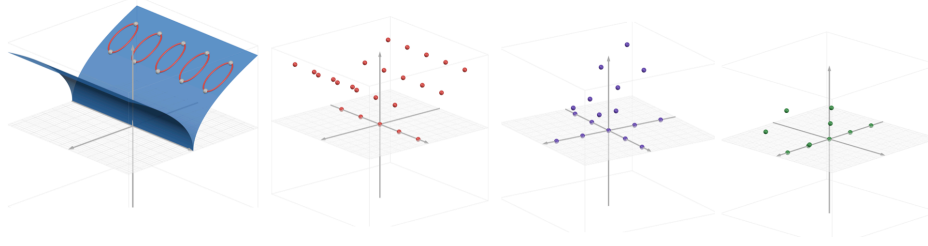
Numerical Irreducible Decomposition. The algebraic decomposition method described in the Preliminaries - Algebraic Background Without Noise subsection of Supplementary Note 3 requires exact symbolic computations and is therefore not stable under coefficient perturbations of the consequences. We address this by replacing symbolic primary decomposition with *numerical irreducible decomposition*, a well-established technique in numerical algebraic geometry [5].

The numerical approach proceeds in two stages. First, the variety is intersected with generic hyperplanes until a zero-dimensional (finite) set of points called witness points is obtained, as illustrated in Supplementary Fig. 2. These witness points sample the variety at discrete locations. Second, homotopy continuation-based curve tracing determines which points lie on the same irreducible components (Supplementary Fig. 3). The method leverages foundational constructs from numerical algebraic geometry, including witness sets [64, 33] and monodromy-based component identification [63]. Tracking points along a homotopy is typically done using predictor-corrector methods along a discretization of the path, using Newton iterations to refine the solutions. Tracking a witness set through multiple homotopies can be used to form a monodromy that leads back to approximations of the original witness set. Each endpoint of the sequence of homotopies is identified with the closest witness point. If two points are identified by a sequence of homotopies, they are path connected and must lie in the same component. This sequence of operations is performed multiple times to produce witness sets for each of the irreducible components of the original variety.

The output of numerical irreducible decomposition is a collection of witness sets - finite point samples on each irreducible component of the variety. While this provides geometric information about the decomposition, it does not directly yield the algebraic structure we seek (polynomial generators of each component). We therefore perform symbolic regression on the witness points to recover candidate polynomial relations.



Supplementary Fig. 2: **Geometric visualization of slicing.** Intersecting generic hyperplanes with a variety until a zero-dimensional set is obtained. Left: the variety. Middle: the slicing process. Right: the zero-dimensional witness set.



Supplementary Fig. 3: **Geometric visualization of curve tracing.** Defining homotopies between points to test whether sets of points lie on the same irreducible surface. Left: illustration of homotopy paths along the surface. Right: sets of points partitioned by irreducible component. Details in [5].

Symbolic Regression on Witness Sets. Given witness points on an irreducible component, we fit polynomial relations that approximately vanish on those points. This problem-learning polynomials from data-has been studied extensively in the context of vanishing ideals [43, 34].

For each removed axiom A_i with symbolic form

$$A_i(x_1, \dots, x_m) = \sum_k c_{ik} M_{ik}(x_1, \dots, x_m) = 0,$$

we use its *monomial support* $\{M_{ik}\}$ as a fixed regression basis. Given sampled points $\{p_j\}_{j=1}^N$ on a single irreducible component, we form the evaluation (Vandermonde-like) matrix $\Phi \in \mathbb{C}^{N \times K}$ with entries $\Phi_{jk} = M_{ik}(p_j)$. The coefficients $\mathbf{c}_i = (c_{i1}, \dots, c_{iK})$ are estimated by solving

$$\min_{\mathbf{c}_i} \|\Phi \mathbf{c}_i\|_2^2 \quad \text{subject to} \quad \|\mathbf{c}_i\|_2 = 1,$$

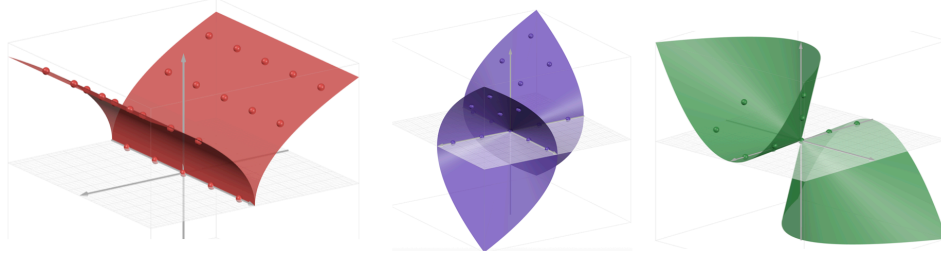
which is equivalent to computing the right singular vector corresponding to the smallest singular value of Φ [43]. This SVD formulation also yields the quality metric used throughout our experiments. The minimum of $\|\Phi \mathbf{c}_i\|_2$ over unit-norm coefficient vectors is exactly $\sigma_{\min}(\Phi)$, attained at the corresponding right singular vector, so $\sigma_{\min}(\Phi)$ directly measures how closely the recovered polynomial vanishes on the witness points: a small value certifies that the sampled points lie nearly within the variety of $\hat{A}_i$, while a value bounded away from zero indicates that no polynomial in the chosen monomial basis approximately vanishes on the sampled points and that the component is therefore not well described by the support of the removed axiom. We caution that a small $\sigma_{\min}(\Phi)$ certifies vanishing on the sampled points but not physical meaningfulness; degenerate axis-aligned branches can also produce very small $\sigma_{\min}$, and these are distinguished from the physical branch by comparison of the recovered coefficients against the monomial support of the missing axiom. We report $\sigma_{\min}(\Phi)$ alongside the recovered polynomial for each component; concrete values for the noisy Kepler case are shown in Scenario 3 of the Examples of AI-Noether's Outputs: Kepler's Third Law

subsection of Supplementary Note 4, where the physical branch achieves $\sigma_{\min} \approx 8.5 \times 10^{-5}$ while the degenerate single-monomial branches collapse to $\sigma_{\min} \sim 10^{-60}$. This formulation identifies the (approximate) kernel of the evaluation map, recovering a polynomial that nearly vanishes on the sample set. Alternative approaches based on border basis methods [34] or least-squares sensitivity analysis could be employed, though we adopt the SVD-based approach for its numerical stability and direct connection to nullspace identification.

The resulting polynomial $\hat{A}_i(\mathbf{x}) = \sum_k \hat{c}_{ik} M_{ik}(\mathbf{x})$ is the best-fitting algebraic relation in the least-squares sense that holds on the numerically computed points. Each component yields its own fitted relation, enabling us to capture branch-specific structures when the noisy variety decomposes into multiple sheets.

Choice of Variables and Basis Functions. The variable set and monomial basis used in regression serve as tunable hyperparameters. For each axiom A_i , we restrict regression to the variables actually appearing in A_i , denoted $\text{vars}(A_i)$, and use its observed monomial support as basis functions. This prevents overfitting and ensures fitted coefficients are directly comparable to the symbolic axiom. Alternatively, one could learn variable subsets or expand the monomial basis to include cross-variable interactions or higher-degree terms, though we adopt the fixed-support approach for clarity and stability.

The fitted residual $\|\Phi \mathbf{c}_i\|_2 / \|\mathbf{c}_i\|_2$ serves as a numerical quality metric. Low residuals indicate that the recovered relation nearly vanishes on the component, suggesting the axiom (or its noisy perturbation) governs that portion of the variety. The resulting polynomials form our set of candidate axioms to test, as illustrated in Supplementary Fig. 4.



Supplementary Fig. 4: **Geometric visualization of symbolic regression output.** Recovered polynomial relations governing the irreducible surfaces identified by curve tracing (Supplementary Fig. 3).

Robust Fitting. In practice, witness sets may contain outliers due to numerical error accumulation in homotopy continuation. To improve robustness, we implement an iterative outlier removal procedure: after initial coefficient estimation, we remove points with the largest residuals (top 10% by default) and refit. This process repeats until convergence or a maximum iteration count, yielding coefficient estimates less sensitive to numerical artifacts. Implementation details, including coefficient normalization and error metrics for comparing recovered and true axioms, are provided in the code repository.

Reason with Axioms and Consequences Containing Numerical Constants.

An alternative to algebraic derivability is to employ a theorem prover to determine whether the conjecture is logically derivable from the expanded set of axioms (i.e., the original axioms together with the abduced ones). Our approach follows the strategy proposed in AI-Descartes [12], which introduced two complementary notions of derivability for scientific laws: a direct (symbolic) derivability test and an *existential* one. The latter is designed for situations in which expressions inferred from data contain numerical constants (e.g., $Q = x/(0.709x + 0.157)$), since such expressions are unlikely to be matched syntactically (often the background theory contains only symbolic constants/parameters). To do this, AI-Descartes abstracts each numeric constant into a new logical variable c_i , for example,

$$Q = \frac{x}{0.709x + 0.157} \rightsquigarrow Q' = \frac{x}{c_1x + c_2},$$

and constrains these variables through admissibility conditions $\mathcal{C}'$ (e.g., non-negativity). The constants are then existentially quantified so that the theorem prover may search over all admissible instantiations.

Under this abstraction, AI-Descartes defines a conjecture to be *existentially derivable* from the background theory when

$$\exists c_1 \dots \exists c_s. (\mathcal{C} \wedge \mathcal{A}) \rightarrow (f' \wedge \mathcal{C}'), \quad (\text{S2})$$

that is, when some choice of the abstracted constants makes the conjecture consistent with both the axioms $\mathcal{A} = \{A_1, \dots, A_{k+r}\}$ and the constraints $\mathcal{C}$ over the symbolic variables present in the axioms ($x_1, \dots, x_n$, assumed universally quantified). This mechanism allows the system to assess the derivability of a data-derived expression with respect to the background theory for at least one admissible instantiation of its numerical constants, thus avoiding sensitivity to noise in their estimated values.

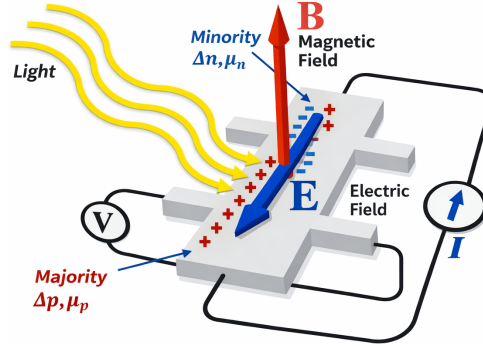
In our setting, some axioms are abduced from the conjecture and the original theory, and thus may also contain numerical constants. Consequently, we need to introduce existential variables not only for the numerical constants appearing in the conjecture but also for those occurring in the abduced axioms. This requires extending the existential derivability notion of Cornelio et al. [12] to handle numerical parameters on both sides of the derivation. Our resulting formulation, which accommodates constants in both the conjecture and the axioms, is as follows:

$$\exists c_1 \dots \exists c_s \left(\mathcal{C}' \wedge \forall x_1 \dots \forall x_n ((\mathcal{C} \wedge \mathcal{A} \wedge \mathcal{A}') \rightarrow f') \right). \quad (\text{S3})$$

where $\mathcal{A}'$ denotes the abduced axioms with their numerical constants replaced by existential variables, and $\mathcal{C}'$ denotes the admissibility constraints for these newly introduced variables.

In this way, we obtain a strict generalization of the existential derivability introduced in AI-Descartes [12], which now accommodates numerical parameters in both the conjecture and the abduced axioms.

Supplementary Note 4: Additional Problem Details and Derivations

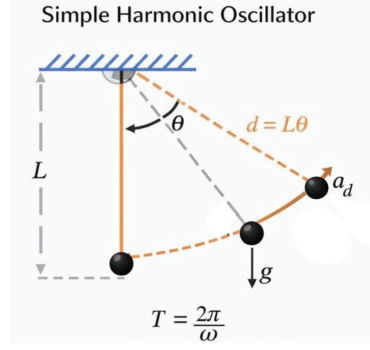


Supplementary Fig. 5: **Illustration of the experimental setup for the Carrier-Resolved Photo-Hall Effect.** Adapted from Gunawan et al. [30], Springer Nature. AI-Noether successfully recovers the necessary Hall coefficient relation required to prove the relationship between the various parameters of a semiconducting surface. $\mathbf{B}$, magnetic field; $\mathbf{E}$, electric field; V , measured voltage; I , applied current (distinct from the light intensity I in the main text); Δn and μ_n , minority-carrier (electron) photo-carrier density and drift mobility; Δp and μ_p , majority-carrier (hole) photo-carrier density and drift mobility (μ_n and μ_p correspond to μ_N and μ_P in the main text).

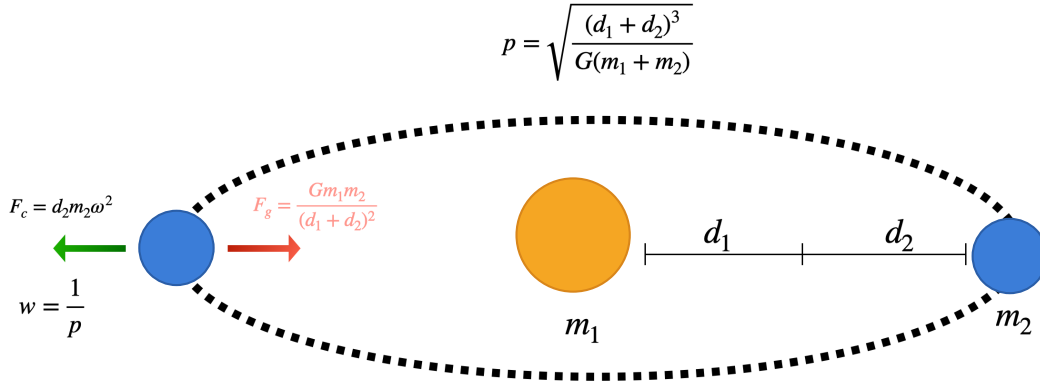
Derivation of Carrier-Resolved Photo-Hall Effect

The Carrier-Resolved Photo-Hall effect [30] equation describes the relationship between various parameters of a semiconducting surface and is given by:

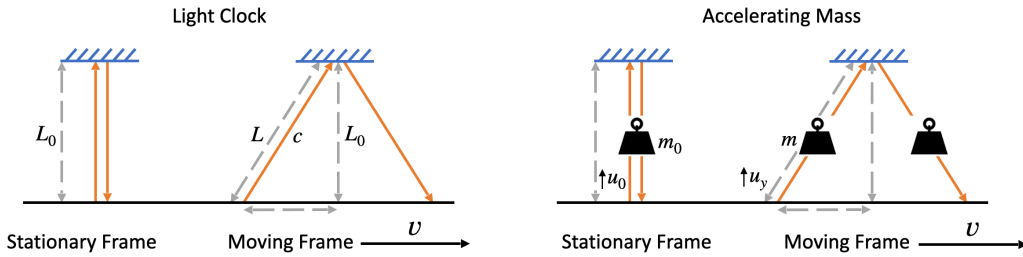
$$Q := H - \frac{r e \mu_P^2 [p_0 + \Delta n(1 - \beta^2)]}{\sigma^2} = 0,$$



Supplementary Fig. 6: **Illustration of the experimental setup for a simple harmonic oscillator.** AI-Noether can infer the requirement of the small angle assumption, gravitational component of acceleration, and any of the other essential axioms required to derive the period of a simple harmonic oscillator. This simpler example illustrates the output of AI-Noether's modules. L , length of the oscillator; θ , angle of displacement; $d = L\theta$, displacement along the arc; a_d , acceleration of the mass along the arc; g , gravitational acceleration; T , period of one oscillation; ω , frequency.



Supplementary Fig. 7: **Illustration of the axioms required to derive Kepler's third law of planetary motion.** AI-Noether detects the existence of gravitational force, an essential axiom required to derive Kepler's third law. When multiple axioms are missing, AI-Noether can also recover coupled versions of the missing axioms. m_1 and m_2 , masses of the two bodies; d_1 and d_2 , their distances from the center of mass; F_c , centrifugal force; F_g , gravitational force; G , gravitational constant; p , orbital period (scaled by a factor of 2π); w , angular frequency (denoted ω in the text).



Supplementary Fig. 8: **Illustration of the multiple experimental setups required to establish each of the relativistic laws - time dilation, length contraction, and relativistic mass.** The left two images illustrate a light clock experiment in a stationary and moving frame of reference required for time dilation and length contraction. The right two images illustrate an accelerated mass experiment in a stationary and moving frame of reference required for deriving relativistic mass. AI-Noether can detect axioms from multiple consequences with more complex derivability dependency structures. Orange arrows denote light paths. L_0 , length of the light clock in the stationary frame; L , length in the moving frame; c , speed of light; v , velocity of the moving frame; m_0 and m , mass in the stationary and moving frames; u_0 and u_y , transverse velocities in the stationary and moving frames.

where $\sigma = e\mu_P[p_0 + \Delta n(1 + \beta)]$ (carrier-resolved Photo-Hall effect [30]). We assume the axioms

$A_1 : \beta\mu_P - \mu_N$	Mobility ratio definition
$A_2 : \mu_H - r\mu$	Hall factor correction
$A_3 : p_h - p_0 - \Delta p$	Hole concentration balance
$A_4 : n - \Delta n$	Electron concentration tracking
$A_5 : \Delta p - \Delta n$	Charge neutrality condition
$A_6 : \sigma - ep_h\mu_P - en\mu_N$	Total conductivity sum
$A_7 : H(p_h + \beta n)^2 e - rp_h + r\beta^2 n$	Hall coefficient relation.

Here μ_N and μ_P are electron and hole drift mobilities, $\beta = \mu_N/\mu_P$ is the mobility ratio (encoded by A_1), μ_H is the Hall mobility with Hall scattering factor $r = \mu_H/\mu$ (encoded by A_2), Δn and Δp are the electron and hole photo-carrier densities (equal in steady state by A_5), H is the Hall coefficient, σ is the conductivity, n is the electron density, p_0 is the background hole density, and p_h is the hole density (for a p-type material).

Solving the coupled axiom system A_1 - A_7 with the measured $\sigma(I)$ and $H(I)$ data simultaneously yields the densities and mobilities of both majority and minority carriers, plus recombination lifetime, diffusion length, and recombination coefficient, providing complete carrier characterization from a single measurement technique for photovoltaic material optimization.

To derive Q , we eliminate $(\mu_N, \Delta p, n, p_h)$ using A_1, A_3 - A_5 , i.e., $\mu_N = \beta\mu_P$, $\Delta p = \Delta n$, $n = \Delta n$, $p_h = p_0 + \Delta n$. Substituting into A_6 gives $\sigma = e\mu_P[p_0 + \Delta n(1 + \beta)]$, and substituting into A_7 gives $eH[p_0 + \Delta n(1 + \beta)]^2 = r[p_0 + \Delta n(1 - \beta^2)]$; replacing $[p_0 + \Delta n(1 + \beta)]^2$ by $\sigma^2/(e^2\mu_P^2)$ yields $Q = 0$. To obtain the polynomial form over measured variables, we multiply the A_7 -derived relation by $e\mu_P$ and use $e\mu_P(p_0 + \Delta n) = \sigma - e\mu_P\beta\Delta n$ from the σ -equation, then expand to get

$$re\mu_P\Delta n\beta^2 + re\mu_P\Delta n\beta - r\sigma + ep_0\sigma H + e\Delta n\beta\sigma H + e\Delta n\sigma H = 0.$$

which is a polynomial version of the Photo-Hall equation. Equivalently, setting

$$\begin{aligned}\alpha_1 &= -\mu e^2\mu_P(\Delta n)^2\beta H - \mu e^2\mu_P p_0\Delta n H - \mu e^2\mu_P(\Delta n)^2 H - \mu e\mu_P p_0 r - \mu\mu_N e\Delta n r + \mu r\sigma, \\ \alpha_2 &= -e\mu_P^2 p_0 - \mu_N e\mu_P \Delta n - e\mu_P^2 \Delta n + \mu_P \sigma, \\ \alpha_3 &= -\mu e^2\mu_P^2 \Delta n\beta H - p_h \mu e^2\mu_P^2 H - \mu e\mu_P^2 \beta r + \mu\mu_N e\mu_P r + \mu e\mu_P^2 r - \mu_H e\mu_P^2, \\ \alpha_4 &= -\mu n e^2\mu_P^2 \beta^2 H - \mu e^2\mu_P^2 \Delta n\beta^2 H - 2p_h \mu e^2\mu_P^2 \beta H + \mu\mu_N e^2\mu_P \Delta n\beta H - \mu e\mu_P^2 \beta^2 r + \mu\mu_N e^2\mu_P p_0 H \\ &\quad + \mu\mu_N e^2\mu_P \Delta n H - \mu\mu_N e\mu_P \beta r + \mu\mu_N^2 e r - \mu_N \mu_H e\mu_P, \\ \alpha_5 &= -\mu e^2\mu_P^2 \Delta n\beta H - p_h \mu e^2\mu_P^2 H - \mu e\mu_P^2 \beta r + \mu\mu_N e\mu_P r + \mu e\mu_P^2 r - \mu_H e\mu_P^2, \\ \alpha_6 &= \mu e\mu_P \Delta n\beta H + \mu e\mu_P p_0 H + \mu e\mu_P \Delta n H - \mu\mu_P \beta r + \mu\mu_N r - \mu_H \mu_P, \\ \alpha_7 &= \mu e\mu_P^2.\end{aligned}$$

we obtain

$$Q = \sum_{i=1}^7 \alpha_i A_i$$

Illustrating Nuances of Axiom Formatting and Multipliers With Kepler's Third Law

From the Multiple Missing Axioms subsection of the Methods in the main text, recall the equation for Kepler's third law, which relates the orbital period of two celestial bodies to their distances and masses. It can be written as

$$Q := p - \sqrt{\frac{(d_1 + d_2)^3}{G(m_1 + m_2)}} = 0,$$

where p is the orbital period (scaled by a factor of 2π), m_1, m_2 the masses, and d_1, d_2 their distances from the center of mass. For circular orbits, we use the following axioms (with p non-

dimensionalized so that $\omega p = 1$):

$$A_1 : (d_1 + d_2)^2 F_g - G m_1 m_2 = 0 \quad (S4)$$

$$A_2 : F_c - m_2 d_2 \omega^2 = 0 \quad (S5)$$

$$A_3 : F_g - F_c = 0 \quad (S6)$$

$$A_4 : \omega p - 1 = 0 \quad (S7)$$

$$A_5 : m_1 d_1 - m_2 d_2 = 0. \quad (S8)$$

Equation (S4) gives gravitational force, (S5) centrifugal force, (S6) equates them, (S7) relates frequency and period, and (S8) defines the center of mass. To derive Q , we eliminate d_1 using (S8), substitute $p = 1/\omega$ from (S7), and replace ω^2 by combining (S5)-(S6) with (S4). This yields

$$G(m_1 + m_2)p^2 - (d_1 + d_2)^3 = 0,$$

which is Kepler's law in polynomial form. Equivalently, setting

$$\alpha_1 = -p^2, \alpha_2 = p^2(d_1 + d_2)^2, \alpha_3 = -p^2(d_1 + d_2)^2, \alpha_4 = m_2 d_2 (\omega p + 1)(d_1 + d_2)^2, \alpha_5 = -d_2^2,$$

we obtain

$$Q = \alpha_1 A_1 + \alpha_2 A_2 + \alpha_3 A_3 + \alpha_4 A_4 + \alpha_5 A_5.$$

Assume that we are missing knowledge of the equation of gravitational force – axiom A_1 –, but we have Kepler's law Q at hand (possibly derived from sufficient data; see [67, 12, 13]). At this point, not only A_1 but also $\alpha_1, \dots, \alpha_5$ are not known. We will assume they exist and attempt to find them from Q and the remaining known axioms $A_2 - A_5$. Here we view $\alpha_1 A_1$ as the (unknown) residual of Q with respect to $A_2, \dots, A_5$. We define the ideal $I = \langle A_2, A_3, A_4, A_5, Q \rangle$. This is the same as the ideal $\langle A_2, A_3, A_4, A_5, \alpha_1 A_1 \rangle$ as $Q = \sum_{i=1}^5 \alpha_i A_i$. Geometrically, $V(A_2, A_3, A_4, A_5, \alpha_1 A_1) = V(A_2, A_3, A_4, A_5) \cap V(\alpha_1 A_1)$. Since the latter is reducible, we have more irreducible components than we have for $V(A_2, A_3, A_4, A_5)$. Computing a primary decomposition of I then gives us the following associated primes:

Decomposition	
Variety 1	$\langle d_2, m_1, F_g, F_c, \omega p - 1 \rangle$
Variety 2	$\langle m_2, d_1, F_g, F_c, \omega p - 1 \rangle$
Variety 3	$\langle m_2, m_1, F_g, F_c, \omega p - 1 \rangle$
Variety 4	$\langle F_c - F_g, m_1 d_1 - m_2 d_2, \omega p - 1, F_g p^2 - m_2 d_2,$ $F_g(d_1 + d_2)^2 - m_1 m_2 G, d_1(d_1 + d_2)^2 - m_2 p^2 G, m_1 p^2 G - d_2(d_1 + d_2)^2,$ $\omega d_1^2(d_1 + d_2)^2 - m_1 p G, \omega d_1^3 - 3\omega d_1 d_2^2 - 2\omega d_2^3 + 2m_1 p G - m_2 p G \rangle$
Generators selected by Reasoning module	
$F_g(d_1 + d_2)^2 - m_1 m_2 G$	

The generators of these irreducible components/ideals represent key equations that define each irreducible component. We iterate through each generator A_1 and compute a Gröbner basis to eliminate the variables not appearing in Q . We find that only the correct expression - $F_g(d_1 + d_2)^2 - m_1 m_2 G$ - can derive Q and is therefore the only candidate returned by our system.

Axiom Formatting and Multipliers. Since the reducibility is introduced by $V(\alpha_i A_i)$, we do not distinguish between α_i and A_i when searching for i th axiom candidates. We may also return axioms in alternative forms; the exact expression is not guaranteed. For example, consider the same Kepler axioms and assume we are missing axiom A_4 instead. Then the relevant ideals in the primary decomposition are:

Decomposition	
Variety 1	$\langle F_c - F_g, m_1d_1 - m_2d_2, \omega p - 1, F_gp - \omega m_2d_2, F_gp^2 - m_2d_2, F_g(d_1+d_2)^2 - m_1m_2G, d_1(d_1+d_2)^2 - m_2p^2G, m_1p^2G - d_2(d_1+d_2)^2, \omega d_2(d_1+d_2)^2 - m_1pG, \omega(d_1 - 2d_2)(d_1+d_2)^2 + 2m_1pG - m_2pG \rangle$
Variety 2	$\langle F_c - F_g, m_1d_1 - m_2d_2, \omega p + 1, F_gp + \omega m_2d_2, F_gp^2 - m_2d_2, F_g(d_1+d_2)^2 - m_1m_2G, d_1(d_1+d_2)^2 - m_2p^2G, m_1p^2G - d_2(d_1+d_2)^2, \omega d_2(d_1+d_2)^2 + m_1pG, \omega(d_1 - 2d_2)(d_1+d_2)^2 - 2m_1pG + m_2pG \rangle$
Generators selected by Reasoning module	
	$\omega p - 1$
	$F_gp^2 - m_2d_2$
	$\omega p + 1$

The highlighted polynomials are returned by our system as being axiom candidates that derive Q along with the remaining axioms. The other generators have been omitted for readability, and cannot derive Q . Although the correct expression $\omega p - 1$ is present, additional terms are present. The term $F_gp^2 - m_2d_2$ is a restatement of A_2 using A_3 and the unknown $\omega p - 1$ substituted in. It contains the same information as $\omega p - 1$ since the former is already known. The term $\omega p + 1$ is an artifact of the fact that α_4 as noted before is $m_2d_2(\omega p + 1)(d_1 + d_2)^2$. Note that the variety $V(m_2d_2(\omega p + 1)(d_1 + d_2)^2)$ is reducible and can be written as the union: $V(m_2) \cup V(d_2) \cup V(\omega p + 1) \cup V((d_1 + d_2)^2)$. Therefore, $\omega p + 1$ appears as a viable axiom candidate since there is an algebraic derivation of Q from the remaining axioms along with $\omega p + 1$. Incidentally, the remaining terms m_2 , d_2 , and $(d_1 + d_2)^2$ also show up in the associated primes:

Decomposition	
Variety 1	$\langle F_c - F_g, m_2^2, (d_1+d_2)^2, m_1d_2 + d_1m_2 + 2m_2d_2, m_1m_2, m_1d_1 - m_2d_2, m_1^2, F_gm_2, F_gm_1, F_g^2, F_g - \omega^2m_2d_2 \rangle$
Variety 2	$\langle d_2, m_1, F_g, F_c \rangle$
Variety 3	$\langle m_2, d_1, F_g, F_c \rangle$
Generators selected by Reasoning module	
	$(d_1+d_2)^2$
	d_2
	m_2

In order to filter out these terms, for non-numerical examples, we apply an additional layer of filtering detailed in the Further Filtering for Projections subsection of Supplementary Note 5.

4.1 Examples of AI-Noether's Outputs: Kepler's Third Law

To illustrate what AI-Noether actually produces at each stage of the *Encode – Decompose – Reason* pipeline, we walk through four scenarios using the Kepler problem defined in `systems_and_phenomena/kepler/system.txt`. All output excerpts below are verbatim from the result tree generated by `python -m src.main --config config.yaml` (the per-run summary files at `results/kepler/abduction/.../summary.txt`).

Indexing convention. The on-disk `system.txt` lists the Kepler axioms in a different order than the narrative of the Multiple Missing Axioms subsection of the Methods in the main text and the preceding subsection. The mapping is:

system.txt index	Polynomial	Paper symbol
1	$m_1 d_1 - m_2 d_2$	A_5 (center of mass)
2	$F_g(d_1 + d_2)^2 - m_1 m_2$	A_1 (gravitational, $G=1$)
3	$F_c - m_2 d_2 \omega^2$	A_2 (centripetal)
4	$F_c - F_g$	A_3 (force balance)
5	$\omega p - 1$	A_4 (Kepler period)

Throughout this subsection we refer to axioms by their `system.txt` index (which is also how the result directories are named: `combo_2` means axiom 2 was removed, `combo_4_5` means axioms 4 and 5 were removed jointly).

Scenario 1: noiseless single-axiom removal (gravitational axiom). We remove axiom 2 ($F_g(d_1 + d_2)^2 - m_1 m_2$) and run the noiseless pipeline. Singular’s `minAss` returns four primary components, and the reasoning stage saves six minimal candidate generator sets. The output reads:

```

=== Abduction Summary ===
Removed axiom indices (1-based): [2]
Removed axioms:
  Fg*(d1+d2)^2 - m1*m2

Decomposition engine: singular
Number of components: 4

Saved combos: 6
['-m1*p^2+d1^2*d2+2*d1*d2^2+d2^3']
['Fg*d1^2+2*Fg*d1*d2+Fg*d2^2-m1*m2']
['m1']
['m2']
['-m1^2*p^2+m1*d2^3+d1*m2*d2^2+2*m2*d2^3',
 '2*m1*p^2+d1^3-3*d1*d2^2-m2*p^2-2*d2^3']
['2*m1*p^2+d1^3-3*d1*d2^2-m2*p^2-2*d2^3',
 'Fg*d1*d2*p+2*Fg*d2^2*p+w*m1*d2^3-m1^2*p']

Strong candidates: 1
['Fg*d1^2+2*Fg*d1*d2+Fg*d2^2-m1*m2']

```

The second saved set, $F_g d_1^2 + 2F_g d_1 d_2 + F_g d_2^2 - m_1 m_2$, is precisely the expanded form of the removed gravitational axiom $F_g(d_1 + d_2)^2 - m_1 m_2$ recovered exactly. The sets $\{m_1\}$ and $\{m_2\}$ are the degenerate sub-varieties discussed in the Further Filtering for Projections subsection of Supplementary Note 5 (setting either mass to zero collapses the system trivially), and the remaining three sets are larger algebraic combinations that include the Kepler target itself.

Scenario 2: noiseless two-axiom removal (case from the Multiple Missing Axioms subsection of the Methods in the main text). We remove axioms 4 and 5 simultaneously - the force balance $F_c = F_g$ (paper’s A_3) and the Kepler period relation $\omega p = 1$ (paper’s A_4) - reproducing the coupled-recovery example already discussed in the Multiple Missing Axioms subsection of the Methods in the main text. This results in the output:

```

=== Abduction Summary ===
Removed axiom indices (1-based): [4, 5]
Removed axioms:
  Fc - Fg
  w*p - 1

Decomposition engine: singular
Number of components: 7

Saved combos: 5
['Fg*p^2-m2*d2']
['-m1*p^2+d1^2*d2+2*d1*d2^2+d2^3']
['m2']

```

```
['m1']
['d1+d2']
```

```
Strong candidates: 1
['Fg*p^2-m2*d2']
```

The first saved set, $F_g p^2 - m_2 d_2$, is the centripetal axiom $A_2 = F_c - m_2 d_2 \omega^2$ with the two missing relations $F_c = F_g$ and $\omega = 1/p$ already substituted in. A *single* recovered polynomial therefore encodes the joint information of both removed axioms - the coupled-recovery phenomenon discussed in the main text. The remaining four saved sets are degenerate $\{m_i\}$, $\{d_1 + d_2\}$, and target-shape sub-varieties.

Scenario 3: noisy single-axiom removal at $\varepsilon = 10^{-2}$. We remove axiom 2 again, but this time supply a noisy target consequence (Gaussian perturbation of magnitude 10^{-2} applied to each coefficient; file `system_noise_1e-2.txt`) and switch to the numerical pipeline. Bertini computes a numerical irreducible decomposition into four witness-set components; per-component symbolic regression then fits each component against the monomial support of the removed axiom $\{F_g d_1^2, F_g d_1 d_2, F_g d_2^2, m_1 m_2\}$. The summary reports:

```
=== Numerical Abduction Summary ===
Noise level: 1e-2
Removed axiom indices (1-based): [2]
Removed axioms:
  Fg*(d1+d2)^2 - m1*m2
```

```
Numerical engine: bertini
Samples per component: 1000
Number of witness set components: 4
Symbolic regression results: 4
```

```
Regression results below threshold (0.5):
Axiom 2, Component 1:
  Singular value: 8.526548500425477e-05
  Polynomial: -0.502463F_g d_1^2 -1.012319F_g d_1d_2
              -0.497537F_gd_2^2 + 0.497536 m_1 m_2
  Coefficient L2 error: 0.021330739164117558
Axiom 2, Component 2:
  Singular value: 6.877503410580804e-94
  Polynomial: 1*Fg*d1^2
  Coefficient L2 error: 1.3228756555322954
Axiom 2, Component 3:
  Singular value: 6.195489941334359e-62
  Polynomial: 1*m1*m2
  Coefficient L2 error: 1.3228756555322954
Axiom 2, Component 4:
  Singular value: 1.2953933564176702e-80
  Polynomial: 1*Fg*d2^2
  Coefficient L2 error: 1.3228756555322954
```

A nuance worth noting: components 2, 3, and 4 each return a *single-monomial* fit (e.g. $F_g d_1^2 = 0$) with an extremely small singular value $\sigma \sim 10^{-60}$ - 10^{-93} . These are *not* good recoveries; they are axis-aligned degenerate branches of the witness variety on which one monomial of the support vanishes identically (the witness points all satisfy $d_1=0$, $d_2=0$, or $m_1 m_2=0$), so σ collapses to numerical zero. Their coefficient L2 error of $\sqrt{1 + (1/\sqrt{2})^2} \approx 1.32$ against the normalized ground-truth axiom is the giveaway. *Component 1* is the physically meaningful branch: with $\sigma \approx 8.5 \times 10^{-5}$ and L2 coefficient error 0.021 ($\approx 2.1\%$), it recovers $-0.496 F_g d_1^2 - F_g d_1 d_2 - 0.491 F_g d_2^2 + 0.491 m_1 m_2 \approx 0$, which after multiplying by 2 is a $\varepsilon = 10^{-2}$ perturbation of the true axiom $F_g (d_1 + d_2)^2 - m_1 m_2 = 0$. The detailed per-component report is then saved.

The recovered axiom is then handed to the Reason stage as a KeYmaera X archive that asks the prover to decide existential derivability (the Handling Noisy Consequences - Numerical Extensions

subsection of Supplementary Note 3; the formulation extends Equation S2 to handle numerical constants in the abduced axioms as well as the consequence) and the abduced axiom's numerical coefficients are abstracted into positive existential variables. The Problem block of the resulting file is:

```

Problem
(
  \exists c1 \exists c2 \exists c3 \exists c4
  \exists c5 \exists c6 \exists c7 \exists c8
  (
    c1>0 & c2>0 & c3>0 & c4>0 & c5>0 & c6>0 & c7>0 & c8>0
    &
    (
      \forall Fc \forall Fg \forall w \forall m1
      \forall d1 \forall m2 \forall d2 \forall p
      (
        (
          Fc>0 & Fg>0 & w>0 & m1>0 & d1>0
          & m2>0 & d2>0 & p>0
          & ( m1*d1-m2*d2 = 0 )
          & ( Fc - m2*d2*w^2 = 0 )
          & ( Fc - Fg = 0 )
          & ( w*p - 1 = 0 )
          & ( c1 * Fg*d1^2 ) = ( c2 * Fg*d1*d2 )
                                + ( c3 * Fg*d2^2 )
                                + ( c4 * m1*m2 )
        )
      ->
      (
        ( c5 * m1*d1*d2^2 ) = ( c6 * m1*m2*p^2 )
                              + ( c7 * d1^2*m2*d2 )
                              + ( c8 * d1*m2*d2^2 )
      )
    )
  )
)
End.

```

Here $c_1, \dots, c_4$ are the existential coefficients of the recovered (Scenario 3, Component 1) axiom, and $c_5, \dots, c_8$ those of the noisy target consequence; the prover decides whether values exist that make the implication hold over the positive orthant.

Scenario 4: a noiseless failure case. Not every removal admits recovery. If we remove axioms 1 and 2 simultaneously - both the center of mass equation $m_1 d_1 = m_2 d_2$ and the gravitational law $F_g(d_1 + d_2)^2 = m_1 m_2$ - the residual axiom system loses all coupling between m_1 and the orbital period, and AI-Noether returns an empty candidate set. The file reads in full:

```

=== Abduction Summary ===
Removed axiom indices (1-based): [1, 2]
Removed axioms:
  m1*d1-m2*d2
  Fg*(d1+d2)^2 - m1*m2

Decomposition engine: singular
Number of components: 1

Saved combos: 0

Strong candidates: 0

```

Singular found a single primary component, but none of the generators (alone or in combination up to `max_candidate_size`) projected onto the target after augmenting the remaining axioms $\{F_c=m_2d_2\omega^2, F_c=F_g, \omega p=1\}$. This is the expected behavior: when too much of the underlying physics is removed at once, no minimal recovery can be both consistent with the target and supported by the surviving axioms, and the system correctly reports failure rather than fabricating a spurious axiom.

Summary. Together, the four scenarios trace the system end-to-end: *Encode* (the `system.txt` axioms and target), *Decompose* (the component counts in each summary, the underlying primary or numerical irreducible decompositions in `decomposition/`), and *Reason* (the minimal saved candidate sets, or the KeYmaera X archive in the noisy case). The on-disk directory layout mirrors this structure; see `results/README.md` for the full output schema.

4.2 Detailed Breakdown of Single Axiom Removal Experiment

We tested the system by removing one axiom at a time and asking it to recover the missing axiom, reporting our findings in Supplementary Table 2. Across 12 problems, the method succeeded in 73 out of 75 cases (97.3%). Recovery was perfect for Kepler (5/5), Time Dilation (5/5), Escape Velocity (5/5), Light Damping (5/5), Hall Effect (7/7), Inelastic Relativistic Collision (9/9), Hagen-Poiseuille (4/4), Neutrino Decay (5/5), Simple Harmonic Oscillator (6/6), Carrier-Resolved Photo-Hall (7/7), and Compton Scattering (10/10). The two failures occurred in the Relativistic Laws system (5/7 recovered), where axioms A_4 (light path geometry) and A_7 (parallel light travel time) were not recovered; these quadratic constraints factor into linear generators during primary decomposition, and the system returns the factored forms rather than the original quadratic axioms (see the Multiple Consequences subsection of the Methods in the main text for discussion).

Supplementary Table 2: Results across problems; axioms are listed per problem as $A_1, A_2, \dots$. “Recovered” indicates whether the method successfully derived the removed axiom.

Problem	Axiom	Expression	Recovered
Hagen-Poiseuille	A1	$u - c_0 - c_2 r^2$	✓
	A2	$\mu \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) - r \frac{dp}{dx}$	✓
	A3	$c_0 + c_2 R^2$	✓
	A4	$L \frac{dp}{dx} = -\Delta p$	✓
Kepler	A1	$(d_1+d_2)^2 F_g - m_1 m_2$	✓
	A2	$F_c - m_2 d_2 \omega^2$	✓
	A3	$F_c - F_g$	✓
	A4	$\omega p - 1$	✓
	A5	$m_1 d_1 - m_2 d_2$	✓
Time Dilation	A1	$c dt_0 - 2d$	✓
	A2	$4L^2 - 4d^2 - v^2 dt^2$	✓
	A3	$f_0 dt_0 - 1$	✓
	A4	$f dt - 1$	✓
	A5	$c dt - 2L$	✓
Escape Velocity	A1	$K_i - \frac{1}{2} m v_e^2$	✓
	A2	$K_f = 0$	✓
	A3	$U_i r + GMm$	✓
	A4	$U_f = 0$	✓
	A5	$K_i + U_i - (K_f + U_f)$	✓
Light Damping	A1	$S r^2 - q_c^2 a_p^2 \sin^2 \theta$	✓
	A2	$dA - 2\pi r^2 \sin \theta d\theta$	✓
	A3	$P - \int_0^\pi S dA$	✓
	A4	$\frac{4}{3} - \int_0^\pi \sin^3 \theta d\theta$	✓
	A5	$a_p^2 - \frac{1}{2} w^4 x_0^2$	✓

Problem	Axiom	Expression	Recovered
Neutrino Decay	A1	$p_\nu - p_\mu$	✓
	A2	$E_\pi - m_\pi$	✓
	A3	$E_\nu - p_\nu$	✓
	A4	$E_\pi - E_\mu - E_\nu$	✓
	A5	$E_\mu^2 - p_\mu^2 - m_\mu^2$	✓
Simple Harmonic Osc.	A1	$a_d - g \sin \theta$	✓
	A2	$d - L\theta$	✓
	A3	$T\omega - 2\pi$	✓
	A4	$d\omega^2 - a_d$	✓
	A5	$T_j - jT$	✓
	A6	$\sin \theta - \theta$	✓
Carrier-Resolved PH	A1	$\beta\mu_P - \mu_N$	✓
	A2	$\mu_H - r\mu$	✓
	A3	$p_h - p_0 - \Delta p$	✓
	A4	$n - \Delta n$	✓
	A5	$\Delta p - \Delta n$	✓
	A6	$\sigma - e p_h \mu_P - e n \mu_N$	✓
	A7	$H(p_h + \beta n)^2 e - r p_h + r \beta^2 n$	✓
Relativistic Laws	A1	$f_0 t_0 - 1$	✓
	A2	$f t - 1$	✓
	A3	$ct_0 - 2L_0$	✓
	A4	$\frac{c^2 t^2}{4} - L_0^2 - \frac{v^2 t^2}{4}$	✗
	A5	$m_0 u_0 - m u_y$	✓
	A6	$u_0 t_0 - u_y t$	✓
	A7	$t(c^2 - v^2) - 2Lc$	✗
Hall Effect	A1	$F_m - q_e v B$	✓
	A2	$F_e - q_e E$	✓
	A3	$F_m - F_e$	✓
	A4	$Eh - U_H$	✓
	A5	$v \, dt - L$	✓
	A6	$I \, dt - Q$	✓
	A7	$Q - N q_e$	✓
Inelastic Relativistic Collision	A1	$p_m^2(c^2 - v_m^2) - m_m^2 v_m^2 c^2$	✓
	A2	$E_m^2 - (m_m c^2)^2 - (p_m c)^2$	✓
	A3	$E_r - m_r c^2$	✓
	A4	$E_c^2 - (m_c c^2)^2 - (p_c c)^2$	✓
	A5	$2E_m E_r - E_c^2 + E_m^2 + E_r^2$	✓
	A6	$p_c - p_m$	✓
	A7	$v_m - \frac{4}{5}c$	✓
	A8	$m_r - m_m$	✓
	A9	$p_c^2(c^2 - v_c^2) - m_c^2 v_c^2 c^2$	✓
Compton Scattering	A1	$E_1 + E_{e,1} - E_2 - E_{e,2}$	✓
	A2	$E_1 - hf_1$	✓
	A3	$E_2 - hf_2$	✓
	A4	$p_1 c - hf_1$	✓
	A5	$p_2 c - hf_2$	✓
	A6	$\lambda_1 f_1 - c$	✓
	A7	$\lambda_2 f_2 - c$	✓

Problem	Axiom	Expression	Recovered
	A8	$E_{e,1} - m_e c^2$	✓
	A9	$E_{e,2}^2 - c^2 p_{e,2}^2 - m_e^2 c^4$	✓
	A10	$p_{e,2}^2 - p_2^2 - p_1^2 + 2p_1 p_2 \cos \theta$	✓

Supplementary Note 5: Additional Technical Details

5.1 Differences Between Real and Complex Solutions

Let $A_1, \dots, A_k$ be a collection of polynomials with $A_1 = 0, \dots, A_k = 0$ being the associated axioms. Let $\mathcal{A}$ stand for the set $\{A_1, \dots, A_k\}$. Let $V_R(\mathcal{A})$, respectively $V_C(\mathcal{A})$, be the set of real, respectively complex, solutions of the equations $A_1 = 0, \dots, A_k = 0$. Note that $V_R(\mathcal{A})$ is the same as the variety $V(\mathcal{A})$ defined in the main body of the paper.

First note that $V_C(\mathcal{A}, Q) = V_C(\mathcal{A})$ if and only if $Q = 0$ for all complex solutions of $A_1 = 0, \dots, A_k = 0$. Further, if $Q \in I(\mathcal{A})$, then $V_C(\mathcal{A}, Q) = V_C(\mathcal{A})$. However, the converse is not true. By Hilbert's Nullstellensatz, if $V_C(\mathcal{A}, Q) = V_C(\mathcal{A})$, then $Q^m \in I(\mathcal{A})$ for some positive integer m (but m need not equal 1). But $Q^m(\bar{x}) = 0$ if and only if $Q(\bar{x}) = 0$. Accordingly, if we say that $Q = 0$ is derivable from $\mathcal{A}$ if and only if $Q^m \in I(\mathcal{A})$, then the notions of “a consequence of” and “derivable from” are identical assuming complex-valued solutions are relevant.

On the other hand, if we are only interested in real-valued solutions, the situation is more complicated. As before, $V_R(\mathcal{A}, Q) = V_R(\mathcal{A})$ if and only if $Q = 0$ for all real solutions of $A_1 = 0, \dots, A_k = 0$. Further, if $Q \in I(\mathcal{A})$, then $V_R(\mathcal{A}, Q) = V_R(\mathcal{A})$. However, the converse is not true. If $V_R(\mathcal{A}, Q) = V_R(\mathcal{A})$, then $Q^{2m} + S \in I(\mathcal{A})$ for some positive integer m and some polynomial S that is a sum of squares of polynomials in R .

5.2 Further Filtering for Projections

While testing ideal membership via Gröbner basis elimination (as described in the Preliminaries - Algebraic Background Without Noise subsection of Supplementary Note 3) establishes that Q is derivable from the candidate axioms together with the known axioms, this criterion alone can admit *trivial* candidates that make the system degenerate. For instance, a candidate axiom that simply sets a variable to zero (e.g., $m_2 = 0$) may technically allow Q to be derived, but only because it collapses the variety onto a lower-dimensional subspace where Q holds vacuously. To filter such cases, we introduce a stronger derivability test based on *exact containment* in the elimination ideal.

Exact Containment Test. Let $\mathcal{A} = \{A_1, \dots, A_k\}$ denote the known axioms, $\hat{A}_{k+1}$ a candidate axiom, and Q the target hypothesis defined over variables $\{x_1, \dots, x_d\}$. We form the augmented ideal $J = \langle A_1, \dots, A_k, \hat{A}_{k+1} \rangle$ and compute its reduced Gröbner basis $\mathcal{G}$ with respect to a lexicographic term order. The elimination ideal $J \cap \mathbb{R}[x_1, \dots, x_d]$ is generated by $\mathcal{G}' = \mathcal{G} \cap \mathbb{R}[x_1, \dots, x_d]$.

Weak test (ideal membership): We accept $\hat{A}_{k+1}$ if $Q \in \langle \mathcal{G}' \rangle$, i.e., if Q reduces to zero modulo $\mathcal{G}'$. This confirms that Q is derivable from the augmented system.

Strong test (exact containment): We accept $\hat{A}_{k+1}$ only if $Q \in \mathcal{G}'$, i.e., if Q appears *exactly* as an element of the Gröbner basis (up to scalar multiplication). This ensures that the projection onto $\text{vars}(Q)$ yields Q directly, rather than a stronger polynomial from which Q follows as a consequence.

Geometric Interpretation. Geometrically, the weak test verifies that $V(\mathcal{A} \cup \{\hat{A}_{k+1}\})$ projects onto $V(Q)$ when restricted to the coordinates of Q . However, this projection may pass through a lower-dimensional intermediate variety. The strong test ensures that the projection is *minimal*: we project directly onto $V(Q)$ without factoring through a smaller-dimensional subvariety.

For example, consider Kepler's third law in polynomial form $Q : G(m_1 + m_2)p^2 - (d_1 + d_2)^3 = 0$, defined over variables $\{p, m_1, m_2, d_1, d_2\}$. A trivial candidate axiom $\hat{A}_{k+1} : m_2 = 0$ would satisfy the weak test, since setting $m_2 = 0$ implies $Gm_1 p^2 - (d_1 + d_2)^3 = 0$, which algebraically contains Q . However, Q itself does not appear in the elimination ideal; instead, we derive a stronger constraint $Gm_1 p^2 - (d_1 + d_2)^3 = 0$ defined over fewer variables (without m_2). The strong test rejects such candidates, filtering out degenerate cases where the system collapses onto boundary conditions.

Limitations and Practical Implementation. The strong test is not foolproof. If the background theory $\mathcal{A}$ already contains an axiom defined over a strict subset of $\text{vars}(Q)$, then the elimination ideal will necessarily include polynomials over fewer variables than Q . For instance, in the Kepler system, the center-of-mass axiom $m_1 d_1 - m_2 d_2 = 0$ is defined over $\{m_1, m_2, d_1, d_2\}$, a proper subset of the variables in Kepler’s third law. Consequently, $\mathcal{G}'$ will contain polynomials over these variables, and Q may not appear exactly in $\mathcal{G}'$ even for legitimate candidate axioms.

To handle this, we adopt a two-tier filtering strategy:

1. **Primary filter (strong test):** Accept candidates where $Q \in \mathcal{G}'$ exactly. These are high-confidence results.
2. **Secondary filter (weak test):** Accept candidates where $Q \in \langle \mathcal{G}' \rangle$ but $Q \notin \mathcal{G}'$, flagging them as lower-confidence. These may include legitimate axioms whose projection is mediated by background constraints.

In our experiments, we save both sets of candidates, noting which satisfy the strong versus weak test. For systems with hierarchical variable dependencies (e.g., conservation laws involving subsets of state variables), the weak test often recovers correct axioms that the strong test would reject.

5.3 Algorithm Implementation Details

Algorithm 1 describes our procedure for identifying missing axioms that render a target hypothesis Q derivable. The algorithm proceeds in three phases that parallel our methodological framework: **Encode**, **Decompose**, and **Reason**, with separate pathways for exact (symbolic) and noisy (numerical) scenarios. We now describe each step in detail.

Input and Initialization. The input consists of a finite set of polynomial axioms $\mathcal{A} = \{A_1, \dots, A_k\}$ in variables $\mathcal{X} = \{x_1, \dots, x_n\}$, a hypothesis polynomial Q over a restricted subset of variables $\{x_1, \dots, x_d\}$ (where $d \leq n$), and a flag *noisy* indicating whether the consequences contain numerical error. The algorithm maintains a set $\mathcal{A}_{\text{expl}}$ of explanatory candidates, initially empty.

Encode. We form the polynomial ideal $I = \langle A_1, \dots, A_k, Q \rangle$. Geometrically, $V(I)$ corresponds to the variety of solutions simultaneously satisfying the known axioms and the hypothesis. If Q is derivable from $\mathcal{A}$, then $V(I)$ coincides with $V(\mathcal{A})$; otherwise, additional structure appears in the decomposition. This encoding step is identical for both exact and noisy cases.

Decompose. The decomposition strategy differs based on whether consequences are exact or noisy:

Exact case (symbolic): We compute a primary decomposition

$$I = I_1 \cap \dots \cap I_r,$$

yielding associated primes $P_j = \sqrt{I_j}$. Each P_j describes an irreducible component of the solution space. The generators of P_j form a finite set $\{\hat{A}_{k+1}^{(j,1)}, \dots, \hat{A}_{k+1}^{(j,m_j)}\}$ of potential candidate axioms.

In our experiments, we use Singular [17], a computer algebra software, for computing the primary decompositions. We also have the functionality for using Macaulay2 [29], another standard computer algebra software, though Singular’s default primary decomposition implementation is more efficient.

Noisy case (numerical): We apply numerical irreducible decomposition [5] to compute witness sets $\{W_1, \dots, W_s\}$, where each W_i is a finite collection of sample points on an irreducible component. For each witness set W_i and each removed axiom $A_\ell \in \mathcal{A}$, we perform symbolic regression: we form the evaluation matrix Φ using the monomial support of A_ℓ evaluated at points in W_i , then compute the right singular vector corresponding to the smallest singular value to obtain fitted coefficients $\hat{c}_\ell$. This yields a candidate axiom $\hat{A}_{k+1}^{(i,\ell)} = \sum_k \hat{c}_{\ell k} M_{\ell k}$ for each component-axiom pair.

We use Bertini [6], an alternate numerical algebraic geometry package with parallelization implementation to compute the witness sets. We also build in the optionality to use Macaulay2’s numerical algebraic geometry package [41].

Reason. For each candidate axiom $\hat{A}_{k+1}$ obtained from decomposition (or sets of candidate axioms if we know that multiple axioms are missing), we test whether it (or they) enable(s) derivation of Q . The reasoning method depends on whether numerical constants are present:

Exact case (algebraic projection): We form the augmented ideal $J = \langle A_1, \dots, A_k, \hat{A}_{k+1} \rangle$ (resp. $\langle A_1, \dots, A_k, \hat{A}_{k+1}, \dots, \hat{A}_{k+l} \rangle$ for testing a set of multiple candidate axioms) and compute its reduced Gröbner basis $\mathcal{G}$ with respect to a lexicographic order on $\mathcal{X}$. We project $\mathcal{G}$ to the coordinate ring of the observed variables: $\mathcal{G}' = \mathcal{G} \cap \mathbb{R}[x_1, \dots, x_d]$. If $Q \in \mathcal{G}'$, then $\hat{A}_{k+1}$ is an *explanatory axiom* for Q and is added to $\mathcal{A}_{\text{expl}}$. The Gröbner basis computations here are done entirely in Macaulay2 [29].

Noisy case (existential reasoning): Following the approach of AI-Descartes [12], we abstract numerical constants in both the hypothesis Q and the candidate axiom $\hat{A}_{k+1}$ (resp. $\hat{A}_{k+1}, \dots, \hat{A}_{k+l}$ for testing a set of multiple candidate axioms) into existentially quantified variables $c_1, \dots, c_s$, yielding Q' and $\hat{A}'_{k+1}$. We encode the derivability query as:

$$\exists c_1 \dots \exists c_s \left(\mathcal{C}' \wedge \forall x_1 \dots \forall x_n \left((\mathcal{C} \wedge \mathcal{A} \wedge \hat{A}'_{k+1}) \rightarrow Q' \right) \right),$$

where $\mathcal{C}$ and $\mathcal{C}'$ denote admissibility constraints on the variables. We submit this query to a theorem prover, KeYmaera X [47] in our implementation. If the prover returns **Proved**, then $\hat{A}_{k+1}$ (or the set of candidates) is added to $\mathcal{A}_{\text{expl}}$.

Output. The algorithm returns the set $\mathcal{A}_{\text{expl}}$ of all candidate axioms that, when added to the background theory $\mathcal{A}$, enable derivation of Q . In practice, we also return metadata including fitted coefficients (for the noisy case), residual norms (quality metrics), and the component indices from which each candidate was extracted.

Computational Complexity. The dominant computational costs are: (i) primary decomposition or witness set computation (doubly exponential in the worst case for symbolic methods [48], polynomial per witness set for numerical methods), (ii) symbolic regression (linear algebra on $N \times K$ matrices for N witness points), and (iii) Gröbner basis computation (doubly exponential worst-case [11], though often tractable in practice for structured systems) or theorem proving. Our experiments demonstrate practical feasibility on systems with up to 16 variables and 10 axioms.

Implementation and Hardware

All symbolic computations were performed using Macaulay2 [29] and Singular [17] for Gröbner basis and primary decomposition and Python/SymPy (for preprocessing and verification). Numerical irreducible decomposition was implemented using Bertini [5, 6] and Macaulay2’s NumericalAlgebraicGeometry package [41]. Symbolic regression on witness sets and theorem proving with KeYmaera X [47] were implemented in Python. Experiments were run on an Apple M4 MacBook Pro with 16 GB of unified memory.

Supplementary Table 3 reports average runtime per test case that ran within the timeout window (7200s) across all benchmark problems, broken down by the number of missing axioms and computational mode (algebraic vs. numerical). Algebraic methods scale efficiently for smaller systems, completing single-axiom tests in under 10 seconds for most problems and maintaining sub-minute performance for systems with up to 10 variables. Larger systems (15-16 variables) require 1-2 minutes per single-axiom test, with moderate increases for multi-axiom cases. Numerical methods incur substantially higher computational costs due to homotopy continuation and witness set construction, typically requiring 10-100x longer than their algebraic counterparts. For the most complex systems (Relativistic Laws, Hall Effect, Compton Scattering), numerical methods require 1-2 hours per test case that succeeded.

This being said, it is also important to note that there were many test cases that failed to run within the allotted time limit of (7200s), especially when 3 axioms were missing. See Supplementary Table 4 for the number of completed runs for each case.

Algebraic methods completed all test cases within the timeout window across all problems and configurations. Numerical methods achieved perfect completion for most problems, but exhibited significant timeouts for two systems. Relativistic Laws completed only 43% (9/21) of single-axiom

Algorithm 1 Identifying Explanatory Axioms via Decomposition and Reasoning

Input: Axioms $\mathcal{A} = \{A_1, \dots, A_k\}$ over variables $\mathcal{X} = \{x_1, \dots, x_n\}$, hypothesis Q over $\{x_1, \dots, x_d\}$, flag *noisy*
Output: Set $\mathcal{A}_{\text{expl}}$ of explanatory axioms

- 1: Initialize $\mathcal{A}_{\text{expl}} \leftarrow \emptyset$
- 2: **Encode:** Form ideal $I \leftarrow \langle A_1, \dots, A_k, Q \rangle$
- 3: **Decompose:**
- 4: **if** *noisy* **then**
- 5: Compute witness sets $\{W_1, \dots, W_s\}$ via numerical irreducible decomposition
- 6: **for** each witness set W_i **do**
- 7: **for** each removed axiom $A_\ell \in \mathcal{A}$ **do**
- 8: Form evaluation matrix Φ from monomial support of A_ℓ on W_i
- 9: Compute $\hat{c}_\ell$ via smallest singular value of Φ
- 10: Construct candidate $\hat{A}_{k+1}^{(i,\ell)} \leftarrow \sum_k \hat{c}_{\ell k} M_{\ell k}$
- 11: **Reason:** Abstract numerical constants in $\hat{A}_{k+1}^{(i,\ell)}$ and Q to obtain $\hat{A}_{k+1}^{(i,\ell)'}$ and Q'
- 12: Formulate existential query: $\exists c_1 \dots \exists c_s (C' \wedge \forall \mathcal{X} ((C \wedge \mathcal{A} \wedge \hat{A}_{k+1}^{(i,\ell)'}) \rightarrow Q'))$
- 13: **if** theorem prover returns *Proved* **then**
- 14: Add $\hat{A}_{k+1}^{(i,\ell)}$ to $\mathcal{A}_{\text{expl}}$
- 15: **end if**
- 16: **end for**
- 17: **end for**
- 18: **else**
- 19: Compute primary decomposition $I = I_1 \cap \dots \cap I_r$
- 20: **for** each associated prime $P_j = \sqrt{I_j}$ **do**
- 21: **for** each generator $\hat{A}_{k+1}^{(j,m)}$ of P_j **do**
- 22: **Reason:** Form ideal $J \leftarrow \langle A_1, \dots, A_k, \hat{A}_{k+1}^{(j,m)} \rangle$
- 23: Compute Gröbner basis $\mathcal{G} \leftarrow \text{GB}(J)$ with lex order $x_1 < \dots < x_n$
- 24: Project: $\mathcal{G}' \leftarrow \mathcal{G} \cap \mathbb{R}[x_1, \dots, x_d]$
- 25: **if** $Q \in \mathcal{G}'$ **then**
- 26: Add $\hat{A}_{k+1}^{(j,m)}$ to $\mathcal{A}_{\text{expl}}$
- 27: **end if**
- 28: **end for**
- 29: **end for**
- 30: **end if**
- 31: **Return** $\mathcal{A}_{\text{expl}}$

cases, 29% (18/63) of pair cases, and 97% (102/105) of triple cases, likely due to the high computational cost of witness set construction for systems with 12 variables and degree-5 polynomials. Compton Scattering experienced severe timeouts, completing only 10% (3/30) of single-axiom cases, 4% (5/135) of pair cases, and less than 1% (2/360) of triple cases - attributable to its large scale (15 variables, 10 axioms, degree 6). Light Damping, having the highest polynomial degree (8), had an increase in runtime as noted in Supplementary Table 3; however, given the manageable number of variables and axioms, it did terminate across all runs.

5.4 Diagnosing Noise Sensitivity in Fig. 6 of the main text

To clarify the non-monotone behavior visible in some of the curves of Fig. 6 of the main text, we isolate AI-Noether’s own noise sensitivity from the noise contributed by the upstream AI-Hilbert step.

Removing AI-Hilbert from the noise chain. In Fig. 6 of the main text, the noise level ε controls the standard deviation of Gaussian perturbations applied to the data columns. AI-Hilbert is then run on the perturbed data to produce a noisy consequence, which is in turn handed to AI-Noether. Hence the noise reaching AI-Noether is filtered through AI-Hilbert’s sensitivity, and this filtering is not in general monotone in ε , since AI-Hilbert’s fit responds to the conditioning of the underlying problem at each noise level. To isolate AI-Noether’s contribution from this upstream step, we re-ran

Problem	Vars	Ax.	Deg.	Algebraic (s)			Numerical (s)		
				1-Ax	2-Ax	3-Ax	1-Ax	2-Ax	3-Ax
Hagen-Poiseuille	9	4	4	0.3	1.1	1.2	3.3	9.5	10.2
Kepler	8	5	4	1.4	6.2	5.8	15.6	30.4	32.5
Time Dilation	8	5	4	4.2	5.9	4.4	20.7	29.3	35.2
Escape Velocity	9	5	4	2.7	3.5	5.4	15.7	26.3	27.2
Light Damping	10	5	8	85.3	106.2	109.9	792.1	1022.2	1032.6
Neutrino Decay	8	5	2	3.4	7.5	9.5	41.2	268.3	251.4
Simple Harmonic Osc.	11	6	5	9.2	13.2	12.5	341.2	690.1	798.3
Carrier-Resolved PH	15	7	6	14.5	59.9	87.2	431.5	1034.2	1242.2
Relativistic Laws	12	7	5	144.2	231.3	202.1	5239.2	6824.4	6523.4
Hall Effect	16	7	4	45.3	104.8	131.1	892.3	3821.9	3733.2
Inelastic Rel. Col.	11	9	6	87.2	163.6	150.2	1222.2	4757.4	5366.3
Compton Scattering	15	10	6	140.1	160.7	172.9	2419.5	6324.2	6823.4

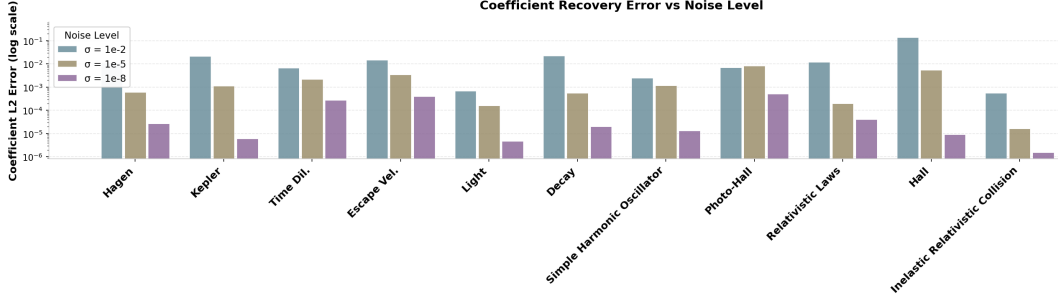
Supplementary Table 3: Average runtime per test case for single, pair, and triple axiom removal experiments. Algebraic computations use primary decomposition and Gröbner basis methods; numerical computations use witness sets, symbolic regression, and an automated theorem prover. Results are averaged over all successful test cases for each problem.

Problem	Vars	Ax.	Deg.	Algebraic			Numerical		
				1-Ax	2-Ax	3-Ax	1-Ax	2-Ax	3-Ax
Hagen-Poiseuille	9	4	4	4/4	6/6	4/4	12/12	18/18	12/12
Kepler	8	5	4	5/5	10/10	10/10	15/15	30/30	30/30
Time Dilation	8	5	4	5/5	10/10	10/10	15/15	30/30	30/30
Escape Velocity	9	5	4	5/5	10/10	10/10	15/15	30/30	30/30
Light Damping	10	5	8	5/5	10/10	10/10	15/15	30/30	30/30
Neutrino Decay	8	5	2	5/5	10/10	10/10	15/15	30/30	30/30
Simple Harmonic Osc.	11	6	5	6/6	15/15	20/20	18/18	45/45	60/60
Carrier-Resolved PH	15	7	6	7/7	21/21	35/35	21/21	63/63	105/105
Relativistic Laws	12	7	5	7/7	21/21	35/35	9/21	18/63	102/105
Hall Effect	16	7	4	7/7	21/21	35/35	21/21	63/63	105/105
Inelastic Rel. Col.	11	9	6	9/9	36/36	84/84	27/27	108/108	252/252
Compton Scattering	15	10	6	10/10	45/45	120/120	3/30	5/135	2/360

Supplementary Table 4: Test case completion rates within the 7200s timeout window. Values show the number of completed runs out of total test cases for each configuration. Algebraic computations use primary decomposition and Gröbner basis methods; numerical computations use witness sets and symbolic regression. The numerical cases are totaled over all three noise levels that were tested on.

each experiment with synthetic Gaussian noise applied directly to the coefficients of the ground-truth consequence (so that no AI-Hilbert step participates in noise propagation), at the same noise levels as in Fig. 6 of the main text. The resulting curves are shown in Supplementary Fig. 9; with the exception of Carrier-Resolved Photo-Hall, the curves are visibly closer to monotone in ε and the worst-case recovery errors are lower than in the AI-Hilbert pipeline experiment.

Omission of Compton Scattering. We omit Compton Scattering from both Fig. 6 of the main text and Supplementary Fig. 9. Under the 7200s timeout, only a small fraction of its noisy runs complete (3/30 single-axiom, 5/135 pair, 2/360 triple; see Supplementary Table 4), so any Compton curve would be averaged over too few runs to support a meaningful comparison, and its apparent non-monotonicity in ε reflects this small sample rather than a property of the recovery method. We therefore exclude it from the plots and report its completion rates in Supplementary Table 4 instead.



Supplementary Fig. 9: **Recovery error when synthetic Gaussian noise is applied directly to the coefficients of the consequence, bypassing AI-Hilbert.** With the exception of Carrier-Resolved Photo-Hall, the curves are closer to monotone in ε than in Fig. 6 of the main text, indicating that the non-monotone behavior in Fig. 6 of the main text is in part attributable to AI-Hilbert’s own sensitivity to the input noise.

Supplementary Note 6: Details of Comparison with State-of-the-Art

We compare our system against the four baselines introduced in the Comparison with State-of-the-Art subsection of the Methods in the main text: (i) Traditional Symbolic Regression (e.g., AI-Feynman [67], Eureqa [20], PySR [15]); (ii) Integrated Symbolic Regression (e.g., AI-Hilbert [13], AI-Descartes [12]); (iii) **LLM**-based abduction using GPT-5 Pro; and (iv) **cvc5**-based abduction. As discussed in the Comparison with State-of-the-Art subsection of the Methods in the main text, the two symbolic-regression baselines (i) and (ii) require observational data for the relationship being fit and therefore cannot operate in our data-free abductive setting: AI-Descartes likewise requires data, while AI-Hilbert run without data in a purely deductive mode fails in every case, since the missing axioms are not deductively derivable from the known axioms and consequences. We therefore benchmark empirically only the two data-free abductive baselines, GPT-5 Pro and cvc5, which we detail below. Both of these baselines receive the same input family of physics systems and targets, and both are evaluated by the same algebraic verifier (Macaulay2) under identical projection checks.

Benchmarks. We use each of the 12 systems from Table 1 of the main text and their noisy counterparts to benchmark the competing methods. For each problem we iterate over all k -tuples of removed axioms with $k \in \{1, 2, 3\}$ (configurable).

LLM

We implement a large language model-based approach to abductive inference using the GPT-5 Pro model (Nov 2025) accessed through the OpenRouter API. For each physics system in our benchmark set, the model is provided with the system’s axioms, the list of measured and non-measured variables (anonymized to avoid introducing biases), and a target polynomial that cannot be derived from the given axioms. We then remove one or more axioms (according to the experimental configuration) and prompt the model to propose an additional axiom or set of axioms whose inclusion would make the target polynomial derivable.

To prevent leakage of semantic cues from variable names and to ensure consistent reasoning across domains, all variables in each system are renamed canonically to $\{x_1, x_2, \dots, x_n\}$. The prompt presents the remaining axioms and the target polynomial in this canonical form and instructs the model to act as an abductive inference solver. The model may reason freely within its response, but is required to conclude with a section formatted as:

Polynomial: < insert one or more candidate polynomials >

Each candidate is required to be distinct from the target polynomial. We parse only the lines following this tag as the model’s proposed abductive hypotheses.

The proposed axioms are automatically translated back into the system’s original variable names using the inverse of the canonical renaming map. For every candidate, we then invoke a symbolic verification step implemented in Macaulay2. The candidate axiom(s) are added to the remaining axioms, and the resulting ideal is projected onto the set of measured variables relevant to the target.

The two checks of *Membership* and *Literal appearance* are then performed corresponding respectively to weak and strong forms of abductive success. The verification procedure is identical to that used for algebraic abduction experiments with Macaulay2.

A proposed axiom set $\mathcal{A}_{\text{abd}}$ is considered *valid* for target Q iff, when added to the given set of axioms $\mathcal{A}_{\text{given}}$, the following hold after eliminating non-measured variables (per target) in Macaulay2:

1. *Membership*: whether the target polynomial reduces to zero modulo the Gröbner basis of the eliminated ideal (i.e., the target lies in the ideal generated by the axioms and proposed axiom). That is, $Q \in \langle \mathcal{A} \rangle$ where $\mathcal{A} = \mathcal{A}_{\text{abd}} \cup \mathcal{A}_{\text{given}}$ (remainder-zero modulo the Gröbner basis of the eliminated ideal), and
2. *Literal appearance (stronger)*: whether the target polynomial appears exactly among the generators of the eliminated Gröbner basis. That is, Q appears *exactly* among the generators of the eliminated ideal’s Gröbner basis.

We report both metrics. For ablations we also track (iii) parsimony (number of added axioms), (iv) solution time, and (v) consistency (the added hypotheses are not already implied by $\mathcal{A}_{\text{given}}$).

All experiments are run using GPT-5 Pro (reasoning model) with temperature 0.7 and a maximum of 1200 tokens per completion. Each combination of system, number of removed axioms, and specific subset of removed axioms is processed independently. For each case, the full prompt, model output, parsed candidate polynomials (in canonical and original variables), and the results of the Macaulay2 verification are written to structured output directories for later analysis. This design enables exact reproducibility of every abductive query and isolates LLM behavior from downstream algebraic validation.

cvc5

CVC5 is a state-of-the-art SMT (Satisfiability Modulo Theories) solver [2] that supports reasoning over a wide range of logical theories, including nonlinear real arithmetic.

We instantiate our setting in CVC5¹ via its native (experimental) *abduct* API, which synthesizes a candidate axiom A_{k+1} such that the asserted axioms $\mathcal{A}_{\text{given}}$ together with A_{k+1} entail the target Q while remaining jointly satisfiable. Data is not required as input, but the axiom has to be exact, that is, without noise (e.g., perturbed hypothesis definition). Both single-axiom and multi-axiom cases are supported. Concretely, we enable abduction (produce-abducts) and query the solver with `getAbduct`. Optionally, by setting `incremental = true` we activate the *incremental mode*, which allows to generate more than one candidate hypothesis axiom via `getAbductNext`. These calls are documented in the official CVC5 Python API.

Each system’s axioms $A_1, \dots, A_k$ and target Q are encoded as quantifier-free nonlinear real arithmetic (QF_NRA): all equations are written in polynomial form, and asserted into the solver. We then pose the abductive query “find A_{k+1} ” with a user-provided grammar restricting the search space of candidate polynomials.

We supply a SyGuS grammar (constructed via `mkGrammar/addRule`) whose start symbol expands to equations of the form $P = 0$, where P ranges over polynomials of bounded total degree. The grammar includes terminals for constants and variables (closed under addition) and non-terminals degree-stratified polynomials (closed under addition, scalar multiplication, and degree-respecting products, e.g., for degree 3 we can have linear $\times$ quadratic). All grammars are generated programmatically from the variable list for each system and capped at a degree inferred from the target (with optional rational coefficient seeds for noisy variants).

As mentioned above we use `getAbduct` together with the grammar to obtain one (or more using `getAbductNext`) candidate A_{k+1} . We then perform two solver-side sanity checks before sending candidates to our Macaulay2 verifier: (i) *Consistency*: $\mathcal{A}$ is satisfiable; and (ii) *Entailment*: $\mathcal{A} \wedge \neg Q$ is unsatisfiable.

To standardize runtimes across problem sizes, we enforce per-query budgets via CVC5’s time and resource limits (e.g., `tlimit-per` in milliseconds). These are set through `setOption` and apply independently to each `abduct` or `consistency` check. For our experiments, we set a limit of 12 hours for each `cvc5` run.

¹All experiments use CVC5 through its Python API (version 1.1.0).

Our results can be summarized in Table 4 of the main text (the Comparison with State-of-the-Art subsection of the Methods).

The single- and multi-axiom recovery rates reported for AI-Noether in Table 4 of the main text are aggregated from the per-problem experiments in the main text, where the specific problems and axioms recovered are listed in full. The 97% single-axiom figure is the 73/75 axioms recovered across all twelve systems, with the complete problem-by-problem and axiom-by-axiom breakdown given in Supplementary Table 2 (the Detailed Breakdown of Single Axiom Removal Experiment subsection of Supplementary Note 4); recovery is exact for every system except the Relativistic Laws, where A_4 and A_7 are instead returned in factored form (see the Multiple Consequences subsection of the Methods in the main text). The 49% multi-axiom figure is the aggregate two-axiom (pair) recovery rate, 105/215 over the same twelve systems, with the per-problem pair and triple counts given in Table 3 of the main text (the Multiple Missing Axioms subsection of the Methods); for reference, three-axiom recovery aggregates to 109/383 ($\approx 28\%$). By contrast, neither cvc5 nor GPT-5 Pro recovered any axiom on these benchmarks.

Summary of Results. Neither baseline successfully recovered missing axioms in our benchmark. The LLM-based approach (GPT-5 Pro) failed to generate any valid candidate axioms across all test cases, producing either syntactically malformed outputs or polynomials that did not satisfy the algebraic verification criteria when back-mapped to the original variable space. The cvc5 solver consistently timed out after 12 hours without producing any abducts, likely due to the combinatorial explosion in the search space induced by the high polynomial degrees (up to degree 8 in our benchmarks) and the number of variables (up to 16). In contrast to traditional symbolic regression methods that require training data for each unknown axiom, and in contrast to these unsuccessful abductive baselines, AI-Noether operates in a purely data-free setting while achieving 97% recovery on single-axiom removal and 49% on multi-axiom removal. This demonstrates that algebraic-geometric decomposition provides a tractable pathway to abductive inference in scientific discovery that remains out of reach for both SMT-based synthesis and LLM-based reasoning under current computational constraints.

References

- [1] Michael F. Atiyah and I. G. Macdonald. *Introduction to Commutative Algebra*, volume 361 of *Addison-Wesley Series in Mathematics*. Addison-Wesley, Reading, MA, 1969. Russian translation published as *Vvedenie v kommutativnuju algebru* (1972).
- [2] Haniel Barbosa, Clark Barrett, Martin Brain, Gereon Kremer, Hanna Lachnitt, Makai Mann, Abdalrhman Mohamed, Mudathir Mohamed, Aina Niemetz, Andres Nötzli, et al. cvc5: A versatile and industrial-strength SMT solver. In *International Conference on Tools and Algorithms for the Construction and Analysis of Systems*, pages 415–442. Springer, 2022.
- [3] Haniel Barbosa, Chantal Keller, Andrew Reynolds, Arjun Viswanathan, Cesare Tinelli, and Clark Barrett. An interactive SMT tactic in Coq using abductive reasoning. In *Proceedings of 24th International Conference on Logic for Programming, Artificial Intelligence and Reasoning (LPAR '23)*, volume 94 of *EPiC Series in Computing*, pages 11–22. EasyChair, 2023.
- [4] Clark Barrett, Roberto Sebastiani, Sanjit A. Seshia, and Cesare Tinelli. *Satisfiability Modulo Theories*, volume 185. IOS Press, 2009.
- [5] Daniel J. Bates, Paul Breiding, Tianran Chen, Jonathan D. Hauenstein, Anton Leykin, and Frank Sottile. Numerical nonlinear algebra, 2024.
- [6] Daniel J. Bates, Jonathan D. Hauenstein, Andrew J. Sommese, and Charles W. Wampler. Bertini: Software for numerical algebraic geometry. Available at bertini.nd.edu with permanent doi: dx.doi.org/10.7274/R0H41PB5.
- [7] Elena Bellodi, Marco Gavanelli, Riccardo Zese, Evelina Lamma, and Fabrizio Riguzzi. Non-ground abductive logic programming with probabilistic integrity constraints. *Theory and Practice of Logic Programming*, 21(5):557–574, 2021.
- [8] Yoshua Bengio. The consciousness prior. *arXiv preprint arXiv:1709.08568*, 2017.

- [9] Daniil A Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578, 2023.
- [10] Steven L Brunton, Joshua L Proctor, and J Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the national academy of sciences*, 113(15):3932–3937, 2016.
- [11] Bruno Buchberger. Bruno buchberger’s phd thesis 1965: An algorithm for finding the basis elements of the residue class ring of a zero dimensional polynomial ideal. *Journal of Symbolic Computation*, 41(3):475–511, 2006. Logic, Mathematics and Computer Science: Interactions in honor of Bruno Buchberger (60th birthday).
- [12] Cristina Cornelio, Sanjeeb Dash, Vernon Austel, Tyler R Josephson, Joao Goncalves, Kenneth L Clarkson, Nimrod Megiddo, Bachir El Khadir, and Lior Horesh. Combining data and theory for derivable scientific discovery with AI-Descartes. *Nature Communications*, 14(1):1777, 2023.
- [13] Ryan Cory-Wright, Cristina Cornelio, Sanjeeb Dash, Bachir El Khadir, and Lior Horesh. Evolving Scientific Discovery by Unifying Data and Background Knowledge with AI Hilbert. *Nature Communications*, 15(1):5922, 2024.
- [14] David Cox, John Little, and Donald O’Shea. *Ideals, Varieties and Algorithms: An Introduction to Computational Algebraic Geometry and Commutative Algebra*. Springer, 1991.
- [15] Miles Cranmer. Interpretable machine learning for science with pysr and symbolicregression.jl, 2023.
- [16] Wang-Zhou Dai, Qiuling Xu, Yang Yu, and Zhi-Hua Zhou. Bridging machine learning and logical reasoning by abductive learning. *Advances in Neural Information Processing Systems*, 32, 2019.
- [17] Wolfram Decker, Gert-Martin Greuel, Gerhard Pfister, and Hans Schönemann. SINGULAR 4-4-0 — A computer algebra system for polynomial computations. <http://www.singular.uni-kl.de>, 2024.
- [18] David Déharbe, Pascal Fontaine, Stephan Merz, and Bruno Woltzenlogel Paleo. Exploiting symmetry in SMT problems. In *Automated Deduction – CADE-22*, volume 5663 of *Lecture Notes in Computer Science*, pages 222–236. Springer, 2009.
- [19] Igor Douven. Abduction. In Edward N. Zalta and Uri Nodelman, editors, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, winter 2025 edition, 2025.
- [20] Renáta Dubčáková. Eureqa: software review. *Genetic Programming and Evolvable Machines*, 12:173–178, 2011.
- [21] Mnacho Echenim, Nicolas Peltier, and Yanis Sellami. A generic framework for implicate generation modulo theories. In *International Joint Conference on Automated Reasoning*, pages 279–294. Springer, 2018.
- [22] Mnacho Echenim, Nicolas Peltier, and Sophie Turrett. An approach to abductive reasoning in equational logic. In *IJCAI 2013-International Joint Conference on Artificial Intelligence*, pages 531–537. AAAI Press, 2013.
- [23] Mnacho Echenim, Nicolas Peltier, and Sophie Turrett. Prime implicate generation in equational logic. *Journal of Artificial Intelligence Research*, 60:827–880, 2017.
- [24] D. Eisenbud. *Commutative Algebra: With a View Toward Algebraic Geometry*. Graduate Texts in Mathematics. Springer, 1995.
- [25] David Eisenbud, Craig Huneke, and Wolmer Vasconcelos. Direct methods for primary decomposition. *Inventiones mathematicae*, 110(1):207–235, Dec 1992.
- [26] Thomas Eiter and Georg Gottlob. The complexity of logic-based abduction. *Journal of the ACM (JACM)*, 42(1):3–42, 1995.

- [27] Han Gao, Luning Sun, and Jian-Xun Wang. Physics-informed deep learning for equation discovery: Methods, challenges, and opportunities. *arXiv preprint arXiv:2304.04862*, 2023.
- [28] Patrizia Gianni, Barry Trager, and Gail Zacharias. Gröbner bases and primary decomposition of polynomial ideals. *Journal of Symbolic Computation*, 6(2):149–167, 1988.
- [29] Daniel R. Grayson and Michael E. Stillman. Macaulay2, a software system for research in algebraic geometry. Available at <http://www2.macaulay2.com>.
- [30] Oki Gunawan, Seong Ryul Pae, Douglas M. Bishop, Yudistira Virgus, Jun Hong Noh, Nam Joong Jeon, Yun Seog Lee, Xiaoyan Shao, Teodor Todorov, David B. Mitzi, and Byungha Shin. Carrier-resolved photo-Hall effect. *Nature*, 575(7781):151–155, November 2019. Epub 2019 Oct 7.
- [31] Boumediene Hamzi and Houman Owhadi. Learning dynamical systems from data: A simple cross-validation perspective, part i: Parametric kernel flows. *Physica D: Nonlinear Phenomena*, 421:132817, 2021.
- [32] Robin Hartshorne. *Algebraic Geometry*, volume 52 of *Graduate Texts in Mathematics*. Springer, 1977.
- [33] Jonathan D. Hauenstein, Jose Israel Rodriguez, and Frank Sottile. General witness sets for numerical algebraic geometry. *Advances in Geometry*, 20(3):381–406, 2020.
- [34] Daniel Heldt, Martin Kreuzer, Sebastian Pokutta, and Hennie Poulisse. Approximate computation of zero-dimensional polynomial ideals. *Journal of Symbolic Computation*, 44(11):1566–1591, 2009.
- [35] Lior Horeish, Kenneth L Clarkson, Cristina Cornelio, and Sara Magliacane. Experimental design for symbolic model discovery, October 28 2021. US Patent App. 16/855,085.
- [36] Yu-Xuan Huang, Wang-Zhou Dai, Le-Wen Cai, Stephen H Muggleton, and Yuan Jiang. Fast abductive learning by similarity-based consistency optimization. *Advances in Neural Information Processing Systems*, 34:26574–26584, 2021.
- [37] Alexey Ignatiev, Nina Narodytska, and Joao Marques-Silva. Abduction-based explanations for machine learning models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 1511–1519, 2019.
- [38] Antonis C. Kakas, Robert Kowalski, and Francesca Toni. Abductive logic programming. *Journal of Logic and Computation*, 2(6):719–770, 1992.
- [39] Hiroaki Kitano. Nobel Turing Challenge: creating the engine for scientific discovery. *npj Systems Biology and Applications*, 7(1):29, 2021.
- [40] P. Langley. Integrated systems for computational scientific discovery. In *Proceedings of the AAAI Conference on Artificial Intelligence 38(20)*, pages 22598–22606, 2024.
- [41] Anton Leykin, Jose Israel Rodriguez, and Frank Sottile. Numerical algebraic geometry for Macaulay2. <https://macaulay2.com/doc/Macaulay2/share/doc/Macaulay2/NumericalAlgebraicGeometry/html/>, 2024. Macaulay2 Package.
- [42] Zachary C Lipton. The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery. *Queue*, 16(3):31–57, 2018.
- [43] Roi Livni, David Lehavi, Sagi Schein, Hila Nachlieli, Shai Shalev-Shwartz, and Amir Globerson. Vanishing component analysis. In Sanjoy Dasgupta and David McAllester, editors, *Proceedings of the 30th International Conference on Machine Learning*, volume 28 of *Proceedings of Machine Learning Research*, pages 597–605, Atlanta, Georgia, USA, 17–19 Jun 2013. PMLR.
- [44] Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024.

- [45] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30, 2017.
- [46] P. Marquis. Extending abduction from propositional to first-order logic. In *In: Jorrand, P., Kelemen, J. (eds) Fundamentals of Artificial Intelligence Research*, 1991.
- [47] Stefan Mitsch and André Platzer. KeYmaera X: An axiomatic tactical theorem prover for hybrid systems. *Journal of Automated Reasoning*, 59(2):175–205, 2017.
- [48] H Michael Möller and Ferdinando Mora. Upper and lower bounds for the degree of gröbner bases. In *International Symposium on Symbolic and Algebraic Manipulation*, pages 172–183. Springer, 1984.
- [49] Rolf Neuhaus. Computation of real radicals of polynomial ideals — II. *Journal of Pure and Applied Algebra*, 124(1):261–280, 1998.
- [50] Peter Norvig. Inference in text understanding. In *AAAI*, pages 561–565, 1987.
- [51] Gabriele Paul. Approaches to abductive reasoning: an overview. *Artificial intelligence review*, 7(2):109–152, 1993.
- [52] Judea Pearl. Reasoning with cause and effect. *AI Magazine*, 23(1):95–95, 2002.
- [53] Charles Sanders Peirce. *Collected papers of charles sanders peirce*, volume 5. Harvard University Press, 1934.
- [54] Brenden K Petersen, Mikel Landajuela Larma, Terrell N. Mundhenk, Claudio Prata Santiago, Soo Kyung Kim, and Joanne Taery Kim. Deep symbolic regression: Recovering mathematical expressions from data via risk-seeking policy gradients. In *International Conference on Learning Representations*, 2021.
- [55] Stanislas Polu and Ilya Sutskever. Generative language modeling for automated theorem proving. In *arXiv preprint arXiv:2009.03393*, 2020.
- [56] Oliver Ray. Automated abduction in scientific discovery. In Lorenzo Magnani and Ping Li, editors, *Model-Based Reasoning in Science, Technology, and Medicine*, volume 64 of *Studies in Computational Intelligence*, pages 103–116. Springer, 2007.
- [57] Andrew Reynolds, Haniel Barbosa, Andres Nötzli, Clark Barrett, and Cesare Tinelli. Scalable algorithms for abduction via enumerative syntax-guided synthesis. In *Automated Reasoning. IJCAR 2020*, volume 12166 of *Lecture Notes in Computer Science*, pages 141–160. Springer, 2020.
- [58] M. T. Ribeiro, S. Singh, and C. Guestrin. ”why should i trust you?”: Explaining the predictions of any classifier. *KDD*, pages 1135—1144, 2016.
- [59] Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M Pawan Kumar, Emilien Dupont, Francisco JR Ruiz, Jordan S Ellenberg, Pengming Wang, Omar Fawzi, et al. Mathematical discoveries from program search with large language models. *Nature*, 625(7995):468–475, 2024.
- [60] Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature machine intelligence*, 1(5):206–215, 2019.
- [61] Parshin Shojaee, Kazem Meidani, Shashank Gupta, Amir Barati Farimani, and Chandan K. Reddy. LLM-SR: Scientific equation discovery via programming with large language models. *arXiv preprint arXiv:2404.18400*, 2024. ICLR 2025 (Oral).
- [62] K.E. Smith, L. Kahanpää, P. Kekäläinen, and W. Traves. *An Invitation to Algebraic Geometry*. Universitext. Springer New York, 2013.
- [63] Andrew J. Sommese, Jan Verschelde, and Charles W. Wampler. Using monodromy to decompose solution sets of polynomial systems into irreducible components. In *Applications of Algebraic Geometry to Coding Theory, Physics and Computation*, volume 36 of *NATO Science Series II: Mathematics, Physics and Chemistry*, pages 297–315. Kluwer Academic Publishers, 2001.

- [64] Andrew J. Sommese, Jan Verschelde, and Charles W. Wampler. Symmetric functions applied to decomposing solution sets of polynomial systems. *SIAM Journal on Numerical Analysis*, 40(6):2026–2046, 2002.
- [65] Simon Spies, Lennard Gäher, Michael Sammler, and Derek Dreyer. Quiver: Guided abductive inference of separation logic specifications in coq. *Proceedings of the ACM on Programming Languages*, 8(PLDI):889–913, 2024.
- [66] D.R. Swanson and N.R. Smalheiser. An interactive system for finding complementary literatures: A stimulus to scientific discovery. *Artificial Intelligence*, pages 183–203, 1997.
- [67] Silviu-Marian Udrescu and Max Tegmark. AI feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16):eaay2631, 2020.
- [68] Mojtaba Valipour, Bowen You, Maysum Panju, and Ali Ghodsi. SymbolicGPT: A generative transformer model for symbolic regression. In *arXiv preprint arXiv:2106.14131*, 2021.
- [69] Zishen Wan, Che-Kai Liu, Hanchen Yang, Chaojian Li, Haoran You, Yonggan Fu, Cheng Wan, Tushar Krishna, Yingyan Lin, and Arijit Raychowdhury. Towards cognitive ai systems: a survey and prospective on neuro-symbolic ai. *arXiv preprint arXiv:2401.01040*, 2024.
- [70] Runxiang Wang, Boxiao Wang, Kai Li, Yifan Zhang, and Jian Cheng. DrSR: LLM based scientific equation discovery with dual reasoning from data and experience. *arXiv preprint arXiv:2506.04282*, 2025.
- [71] Kaiyu Yang, Aidan Swope, Alex Gu, Rahul Chalamala, Peiyang Song, Shixing Yu, Saad Godil, Ryan J Prenger, and Animashree Anandkumar. Leandojo: Theorem proving with retrieval-augmented language models. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 21573–21612. Curran Associates, Inc., 2023.
- [72] Xiao-Wen Yang, Wen-Da Wei, Jie-Jing Shao, Yu-Feng Li, and Zhi-Hua Zhou. Analysis for abductive learning and neural-symbolic reasoning shortcuts. In *Forty-first International Conference on Machine Learning*, 2024.
- [73] Zhi-Hua Zhou. Abductive learning: towards bridging machine learning and logical reasoning. *Science China. Information Sciences*, 62(7):76101, 2019.